
Long-Horizon Audio-Visual Generation for Persistent Stories and Interactive Worlds

Abstract

Video generation is progressing beyond isolated clips toward long-form narratives and interactive worlds, requiring models to preserve identities, follow user controls, and remain stable over extended rollouts. We present **JoyAI-Echo-1.5**, a unified audio-visual generation system with two purpose-built variants. The long-video variant introduces composable cross-shot memory that aggregates visual evidence across multiple prior shots and speaker cues derived from speech-filtered full-shot audio, enabling persistent character appearance and voice identity across flexible combinations of text, image, and memory conditioning. The world-model variant converts heterogeneous navigation inputs into calibrated metric 6-DoF camera trajectories and injects them through a geometry-aware conditioning pathway, enabling controller-agnostic interaction across flexible viewpoints. To support efficient long-horizon generation, we transform a bidirectional audio-visual backbone into a causal few-step generator using progressive teacher forcing and short- and long-horizon Self-Gradient Forcing on self-generated rollouts. Experiments demonstrate strong performance in both settings. JoyAI-Echo-1.5 achieves improvements over existing long-video baselines in cross-shot consistency, visual quality, text alignment, and speech fidelity. Its world-model variant ranks **first** on WBench, with an average score of 81.7, and achieves leading visual quality and long-horizon persistence on SANA-WM-Bench. Together, these results indicate that memory, geometric control, and rollout-aware training provide a practical foundation for generating coherent stories and continuously evolving interactive worlds. Project page: <https://echo-team-joy-future-academy-jd.github.io/Echo-1.5-Page/>.

1 Introduction

The next frontier in video generation is not merely improving individual clips, but achieving **continuity and control**. Long-form creation requires characters and voices to remain consistent across shots, while interactive world modeling requires environments to respond precisely to user intent and remain stable during continuous rollouts. Yet existing video models are still largely designed around fixed temporal windows: they forget identities, drift when conditioned on self-generated histories, and struggle to translate heterogeneous action inputs into consistent motion. Overcoming these limitations requires more than scaling model capacity; it requires rethinking how models remember, act, and evolve over time.

We present **JoyAI-Echo-1.5**, a technical system comprising two purpose-built versions. Its **long-video version** targets long-form audio-visual generation and introduces composable cross-shot memory to preserve character appearance and speaker identity across changing scenes. It aggregates visual evidence from multiple prior shots, derives robust voice cues from speech-filtered full-shot audio, and supports flexible combinations of text, image, and memory conditions. Its **world-model version** targets interactive world modeling by converting heterogeneous navigation inputs into calibrated metric 6-DoF camera trajectories and injecting them through a geometry-aware control pathway.

The first extends stories beyond individual shots; the second transforms generated video into an environment that can be explored.

Both versions are designed to operate beyond fixed temporal windows. JoyAI-Echo-1.5 progressively converts a bidirectional audio-visual backbone into a causal few-step generator through audio-visual teacher forcing followed by short- and long-horizon Self-Gradient Forcing on self-generated rollouts. Training on model-generated histories exposes the system to accumulated generation errors, helping it preserve synchronized audio-visual dynamics and sustain long streaming sequences within a bounded computational budget. **The long-video version gives generation memory. The world-model version gives it control. Together, JoyAI-Echo-1.5 pushes video generation from creating isolated moments toward sustaining stories and evolving worlds.**

Main contributions. Our main contributions are summarized as follows:

- **A long-video system that remembers.** We introduce multi-shot audio-visual memory with speech-filtered voice representations, separated positional regions, and composable text, image, and memory conditioning to preserve identity across shots.
- **A world model that responds.** We establish a controller-agnostic action interface based on calibrated metric 6-DoF trajectories, enabling precise camera control across heterogeneous data sources, environments, and viewpoints.
- **A generator built to continue.** We develop a causal few-step training pipeline that combines audio-visual teacher forcing with short- and long-horizon Self-Gradient Forcing, allowing JoyAI-Echo-1.5 to learn from its own rollout distribution and remain stable over extended generation.

2 Data

2.1 Long-Video Training Data

JoyAI-Echo-1.5 adopts a two-stage data curriculum. **Stage I** reuses the identity-centric corpus of JoyAI-Echo-1.0 [Li et al., 2026] and constructs memory–target pairs from scene-disjoint clips of the same recurring character to train audio-visual memory conditioning. **Stage II** uses high-quality videos with broader resolution and aspect-ratio coverage, increases the proportion of subtitle-free and Chinese-language samples, and incorporates captioned single-shot and multi-shot videos, with the latter explicitly describing transitions across cuts. It improves general audio-visual quality, spatial and linguistic coverage, and transition modeling, while preserving text-to-audio-video (T2AV) and image-to-audio-video (I2AV) generation capabilities. Both stages use the same quality-control pipeline, with their sampling distributions and caption formats adapted to their respective objectives.

2.1.1 Stage-I Memory-Construction Corpus

Identity-centric source groups. The memory-construction corpus retains the identity-centric organization introduced in JoyAI-Echo-1.0 [Li et al., 2026]. Starting from long-form films, television episodes, and web videos, recurring characters are associated with scene-disjoint single-shot clips. Following the extraction pipeline of v1.0, the data undergo global identity discovery, scene grouping, local tracklet assignment, and quality filtering. Consequently, the basic unit used by v1.5 is not an isolated video clip, but an identity group

$$\mathcal{G}_k = \{S_{k,n}\}_{n=1}^{N_k}, \quad S_{k,n} = (V_{k,n}, A_{k,n}), \quad (1)$$

where all clips in $\mathcal{G}_k$ contain the same recurring character while differing in scene, clothing, viewpoint, pose, motion, illumination, facial expression, dialogue, and acoustic environment. Maintaining this diversity is important because a useful memory representation should capture persistent character-level evidence rather than scene-specific appearance alone.

Memory-target sample construction. At each training iteration, one clip $S_{k,t}$ is selected as the prediction target and a variable-size subset of the remaining clips is sampled as memory:

$$\mathcal{M}_{k,t} = \{S_{k,i_1}, S_{k,i_2}, \dots, S_{k,i_K}\}, \quad i_j \neq t. \quad (2)$$

The memory and target clips therefore show the same identity through different observations rather than through temporally adjacent frames from the same trajectory. This separation discourages direct

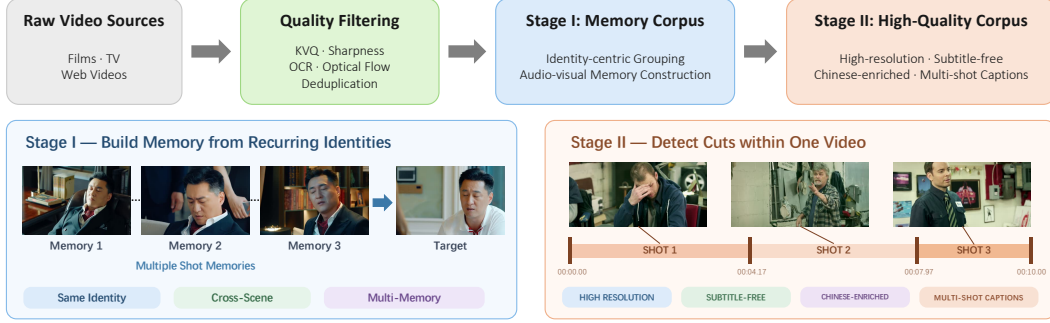


Figure 1: Two-stage data construction pipeline of JoyAI-Echo-1.5. Raw videos are quality-filtered and organized into an identity-centric Memory corpus and a high-quality corpus with broader resolution, language, and multi-shot coverage.

content copying and requires the model to recover stable identity information under changes in background, camera pose, wardrobe, action, and speech content. The number of sampled memory shots K is varied during training so that the model does not rely on a fixed memory capacity and remains effective when only a small amount of character evidence is available.

For every selected memory shot, one video frame is randomly sampled and encoded as visual memory. In contrast to the visual branch, which uses a compact frame-level observation, the complete shot audio is retained. The audio is first processed by the speech-filtering operator and is then encoded as audio memory. Using the complete speech-bearing interval provides a more reliable description of speaker characteristics than selecting a short response window, while speech filtering reduces the leakage of background music and environmental effects into the speaker condition. The visual and audio observations from each memory shot preserve the same slot order, allowing the model to jointly access the appearance and voice evidence associated with that shot.

Conditioning configurations. The availability of memory and the first-frame image condition is independently varied when constructing training samples. This produces four conditioning configurations under a unified interface:

- **T2AV**: text-to-audio-video generation without memory or an input image;
- **I2AV**: audio-video generation conditioned on text and the first frame;
- **MT2AV**: text-to-audio-video generation conditioned on audio-visual memory;
- **MTI2AV**: audio-video generation jointly conditioned on memory, text, and the first frame.

The memory-construction corpus primarily supplies MT2AV and MTI2AV examples, while the same sample representation also supports T2AV and I2AV when the corresponding conditions are absent. Randomizing both memory length and image-condition availability prevents the model from binding a particular generation task to a fixed input layout. It also enables memory and the first-frame condition to be used compositionally, rather than treating memory-conditioned generation and image-conditioned generation as separate models.

2.1.2 Stage-II High-Quality and Multi-Shot Enhancement Corpus

The identity-organized corpus provides the supervision required for cross-shot memory retrieval, but its distribution is constrained by the availability of recurring characters and scene-disjoint observations. Training exclusively on such samples can reduce the diversity and standalone quality of ordinary T2AV and I2AV generation. Stage II therefore uses an enhancement corpus selected independently of the identity-group requirement. It combines high-quality single-shot videos with samples containing multiple shots and explicit transitions. Its purpose is to preserve and strengthen general audio-visual generation, broaden spatial and linguistic coverage, and improve transition modeling after the memory interface has been established in Stage I.

Resolution and aspect-ratio coverage. The high-quality corpus contains videos at 720×720 , 1280×720 , and 720×1280 , corresponding to square (1:1), landscape (16:9), and portrait (9:16)

formats. Rather than concentrating training on one canonical spatial shape, we explicitly balance these formats so that the model encounters comparable high-quality examples across common generation settings. This improves robustness to changes in token layout and spatial composition and allows the same model to serve cinematic landscape generation, portrait-oriented content, and square social-media formats.

Subtitle-free visual data. The retained samples emphasize clear visual content and stable audio-visual quality, with particular attention to videos without embedded subtitles. Burned-in subtitles create persistent, high-contrast patterns that are strongly correlated with dialogue. When such patterns are overrepresented, the model may reproduce incomplete glyphs, garbled character fragments, or subtitle-like regions even when no text is requested. Increasing the proportion of subtitle-free video weakens this undesirable correlation and provides cleaner supervision for faces, body motion, lower-frame image regions, and speaking scenes. OCR metadata is still retained for filtering and analysis, but samples with dominant or repeatedly occluding text are excluded from the enhancement corpus.

Chinese-language expansion. We additionally increase the proportion of Chinese-language data, including Chinese captions and dialogue-oriented audio-visual content. The broader language distribution supplies direct supervision for Chinese prompt following, spoken-content generation, and the alignment between visible speaking motion and Chinese dialogue. This adjustment is particularly important for long-form generation, where errors in linguistic conditioning or speaker behavior can accumulate across shots. The language expansion is performed together with visual-quality and subtitle filtering so that improved Chinese coverage does not come at the cost of image cleanliness or acoustic fidelity.

2.1.3 Quality Filtering and Distribution Balancing

Both corpora inherit the multi-axis quality-control pipeline of JoyAI-Echo-1.0, but the retained distribution is recalibrated for high-resolution and multi-shot training.

Learned perceptual quality. We first apply a learned video-quality estimator [Lu et al., 2024] to assess perceptual quality over both spatial content and temporal behavior. It removes samples with severe compression artifacts, abnormal color, unstable exposure, visually distracting noise, or consistently low aesthetic quality. This learned score complements the following hand-crafted measurements: it captures degradations that are difficult to describe with one low-level statistic, whereas the explicit operators provide interpretable control over identity visibility, text overlays, and motion coverage.

Structural sharpness. Uniformly sampled frames are evaluated using the variance of the Laplacian. For a sampled frame I_t , its sharpness response is

$$s_t^{\text{sharp}} = \text{Var}(\nabla^2 I_t), \quad (3)$$

and the frame responses are robustly aggregated into a clip-level score

$$S_{\text{sharp}}(V) = \text{Agg}\left(\{s_t^{\text{sharp}}\}_{t \in \mathcal{T}}\right), \quad (4)$$

where $\mathcal{T}$ denotes the uniformly sampled timestamps. Unlike face-detection confidence alone, this operator directly measures whether structural details remain visible. It is therefore particularly important for memory training: a face may still be detected under defocus, motion blur, or low spatial resolution, while providing unreliable identity supervision. Clips with persistently weak responses or insufficiently clear character observations are removed.

Text and overlay filtering. OCR and overlay detectors identify subtitles, logos, watermarks, large text regions, and burned-in graphics. We consider not only detected text content but also its occupied area, temporal persistence, and overlap with the main character. Clips in which text dominates the image or repeatedly occludes faces and bodies are rejected. For the high-quality enhancement corpus, this filter is made deliberately stricter to increase the proportion of subtitle-free data. OCR metadata is nevertheless preserved for caption construction: an empty result is explicitly verbalized as no on-screen text or subtitles, whereas visible text is introduced with an OCR marker.

Motion statistics. Dense optical flow [Farneback, 2003] is used to estimate whole-frame dynamics. Given the flow field $\mathbf{u}_t(p)$ between adjacent frames, we compute

$$M(V) = \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{|\Omega|} \sum_{p \in \Omega} \|\mathbf{u}_t(p)\|_2, \quad (5)$$

where Ω is the image domain. The motion score is not used to retain only high-motion samples; instead, it supports distribution control. Very low-motion clips are downsampled to prevent the corpus from being dominated by static talking heads, while sharpness and perceptual quality filters prevent large flow caused by blur, camera shake, or corrupted frames from being mistaken for useful motion. The retained data therefore covers static dialogue, subtle facial and body motion, camera movement, and larger character displacement without collapsing toward either motion extreme.

Identity-level coverage and de-duplication. For the memory-construction corpus, all of the above filters are applied at both clip and identity-group levels. If an identity no longer contains enough scene-disjoint high-quality clips after filtering, the complete group is removed because it cannot support a meaningful memory–target split. Within each identity, perceptual hashes, color histograms, and DINOv2 background similarity are used to suppress near-duplicate observations. This ensures that adding memory slots introduces new evidence about appearance, viewpoint, scene, or voice rather than repeated copies of the same shot.

Distribution balancing. Finally, the retained samples are balanced across square, landscape, and portrait formats; Chinese and English content; subtitle presence; and motion ranges. The identity-organized and high-quality corpora are also mixed so that memory-conditioned examples do not overwhelm ordinary T2AV and I2AV training. These controls prevent one source domain or generation mode from dominating optimization and allow improvements in composable memory conditioning to be obtained without sacrificing general visual and acoustic fidelity.

2.1.4 Single- and Multi-Shot Audio-Visual Captioning

JoyAI-Echo-1.5 expands the clip-level audio-visual annotation of v1.0 into bilingual captions covering both single-shot and multi-shot videos. For each structure, we construct a text-to-video caption and derive a corresponding image-to-video caption. The T2V caption must independently specify the visible subjects and their temporal behavior, whereas the I2V caption can rely on the input first frame for appearance information and is therefore rewritten to reduce redundant identity descriptions.

Single-shot T2V captions. For a single-shot clip, structured fields are concatenated in the following order:

$$c_{\text{single}}^{\text{T2V}} = R \oplus A \oplus S \oplus C \oplus B \oplus O \oplus E \oplus G, \quad (6)$$

where R is Roles_and_Subjects, A is Action_and_Dialogue, S is Style, C is Camera_Movement, B is Background, O is the subtitle/OCR description, E is Sound_Effects, and G is BGM. The action field describes actions, expressions, body movement, and dialogue in temporal order, while voice timbre and speaking manner are retained in the subject and dialogue descriptions. When no OCR or subtitle is detected, O explicitly states that no on-screen text or subtitles are present; otherwise, the recognized content is introduced by an OCR marker. Repeated empty-text statements are removed during concatenation.

Multi-shot T2V captions. For a video containing N shots, we use the rewritten identity-consistency annotation and serialize the individual shot records in chronological order:

$$c_{\text{multi}}^{\text{T2V}} = H_N \oplus \bigoplus_{s=1}^N [Q_s \oplus R_s \oplus A_s \oplus C_s \oplus O_s \oplus E_s \oplus G_s], \quad (7)$$

where H_N states the number of shots and Q_s is the explicit Shot s boundary. Each shot preserves its own subjects, actions and dialogue, camera behavior, OCR state, sound effects, and background music. This format provides direct supervision for shot order, subject changes, dialogue allocation, camera changes, and soundtrack continuity across cuts. Characters recurring across shots retain consistent identifiers in the T2V annotation, while characters appearing for the first time are fully introduced in the shot where they enter.

Table 1: Representative abbreviated examples of the four caption formats used in JoyAI-Echo-1.5. English renderings are shown for readability, while both Chinese and English captions are retained during training. T2V captions preserve explicit identity definitions, whereas I2V captions use concise natural references because appearance is supplied by the first frame.

Structure	T2V caption	I2V caption
Single shot	ID_A is a young man with short brown hair, wearing a dark gray hoodie, with a clear mid-range voice. He sits slightly off-camera, blinks naturally, and says, “my games often focus on that ...” A static chest-up close-up uses shallow depth of field in an indoor gym. No on-screen text or subtitles. No obvious sound effects. No background music.	<i>[Random I2V prefix]</i> A short-haired man speaks with a clear, stable mid-range voice and a natural conversational tone. He sits slightly off-camera, blinks naturally, and says, “my games often focus on that ...” A static chest-up close-up keeps him in focus against a softly blurred gym. No on-screen text or subtitles. No obvious sound effects. No background music.
Multi-shot	This video contains two shots. Shot 1: ID_A is a middle-aged man in a dark green dress uniform with a deep voice; ID_B is a middle-aged man in digital camouflage with a bright voice. They walk and talk side by side as a medium shot follows them. OCR: military slogans and a name label. Soft piano music plays. Shot 2: ID_C is a silent middle-aged man in a dark blue police uniform. A police formation advances in a static medium shot. No OCR or subtitles; the piano music continues.	<i>[Random I2V prefix]</i> This video contains two shots. Shot 1: A middle-aged man in a dress uniform and a middle-aged man in camouflage walk and talk side by side. The former has a deep, steady voice and the latter a bright, natural voice. A medium shot follows them; soft piano music plays. Shot 2: A serious middle-aged man in a dark blue police uniform appears silently within an advancing formation. A static medium shot follows the group. No OCR or subtitles; the piano music continues.

I2V caption rewriting. Both single-shot and multi-shot T2V captions are converted into I2V captions. Because the first frame already provides the initial appearance, the full identifier-based subject description in the first shot is replaced by a concise semi-referential phrase, such as “the short-haired man” or “the uniformed middle-aged man.” The synthetic identifiers such as ID_A are removed. Voice timbre, speaking manner, dialogue, sound effects, and background music are retained because they cannot be recovered reliably from the image. For multi-shot videos, any new character appearing after the first shot receives a sufficiently detailed natural-language description at first appearance, but is likewise described without an identifier token. The complete caption is then lightly shortened to remove repeated visual facts without discarding temporal actions or audio information.

A bilingual I2V instruction prefix is randomly sampled and prepended to the rewritten caption. The same procedure is applied independently to the Chinese and English annotations, and the resulting strings are stored in `video_audio_i2v_caption`. This design reduces conflict between the image and redundant textual appearance descriptions while preserving the information required to generate motion, dialogue, voice, sound, and multi-shot evolution.

The two-stage curriculum separates the acquisition of memory conditioning from the subsequent enhancement of general quality and transition modeling. Stage I teaches the model to retrieve and compose cross-shot identity evidence. Stage II uses high-quality subtitle-free data with increased Chinese coverage to maintain standalone T2AV and I2AV fidelity; within the same stage, captioned single-shot samples preserve fine-grained intra-shot actions and multi-shot samples provide explicit supervision for transitions across cuts. This progression allows JoyAI-Echo-1.5 to extend long-form controllability and composable memory conditioning while maintaining broad spatial, linguistic, visual, and acoustic coverage.

2.2 Action Annotation and Trajectory Data

Action-conditioned world generation requires a consistent motion representation across data collected from different sources. However, these sources provide substantially different levels of supervision: synthetic UE renders expose exact camera poses and control inputs, internal gameplay provides action logs but not camera poses, while human gameplay and web videos provide audio-visual observations but no explicit action logs or camera telemetry. We therefore build a unified processing pipeline that converts all sources into metric 6-DoF camera trajectories and, whenever available, aligns them with source-native control signals.

Trajectory Data from Heterogeneous Sources. We collect motion supervision from three complementary data sources:

- *Unreal Engine (UE) Renders.* UE environments provide accurate geometric supervision. Avatars are controlled through local navigation commands under the engine’s native physics and collision system. Each rollout is rendered from one first-person and four character-relative third-person cameras. For every frame, we synchronously record RGB observations, keyboard inputs, camera intrinsics, and ground-truth camera-to-world poses in metric units. During training, one viewpoint is randomly sampled from each sequence, encouraging the model to generalize across camera configurations.
- *Internal Gameplay.* Internal game environments provide diverse interaction scenarios, including open-world navigation, driving, and mounted traversal in both first- and third-person views. Scripted agents execute predefined control sequences, allowing us to record accurately aligned action commands together with RGB video, stereo audio, and scene metadata. Interface elements such as HUDs and crosshairs are removed during rendering. Since these game engines do not expose camera pose matrices, their 6-DoF trajectories are estimated from the recorded videos and temporally aligned with the corresponding action logs.
- *Human Gameplay and Web Video.* These videos complement scripted data with natural navigation behavior, human reaction patterns, and richer audio-visual content. They do not contain explicit action logs or camera telemetry, so we use the recovered camera trajectory itself as motion supervision. We extract temporally continuous, shot-consistent segments of approximately one minute and estimate their metric 6-DoF camera motion using geometry-based reconstruction.

Together, these sources provide a spectrum from precise synthetic supervision to large-scale natural motion, while sharing the same trajectory representation.

Long-Sequence Metric Pose Recovery. For videos without ground-truth camera poses, directly estimating motion on short training clips often leads to scale ambiguity and unstable trajectories. We instead estimate camera motion over longer, continuous windows of approximately one minute and only afterward divide them into training clips.

Specifically, we apply the metric-depth-assisted ViPE pipeline [Huang et al., 2025] to temporally subsampled frames. The longer temporal context provides a larger geometric baseline and improves the robustness of pose reconstruction. We then recover frame-rate trajectories by interpolating rotations with Spherical Linear Interpolation (SLERP) and translations with linear interpolation. Finally, both estimated and ground-truth trajectories are transformed into a common camera-to-world coordinate convention, metric scale, axis orientation, and temporal sampling rate.

Trajectory Quality Filtering. We apply trajectory-level filtering before training. Clips are removed when pose estimation has low confidence or exhibits implausible motion, including severe frame-to-frame jitter, abrupt rotational changes, negligible displacement, or unstable camera intrinsics. Ground-truth UE trajectories are subjected to the same motion-range checks, while their original metric poses are kept unchanged.

Action-Decoupled Text Annotation. We structure each audio-visual caption into persistent scene attributes, a dynamic-motion narrative, and audio-related descriptions. Persistent attributes include the scene, visual style, viewpoint, and subject identity, while the audio component describes speech and environmental sounds. During Action-SFT, we omit both the dynamic-motion narrative and the audio-related fields, retaining only the persistent static descriptors because this stage uses a visual-only denoising objective. During Joint-FT, we restore the audio-related fields while continuing to exclude the dynamic-motion narrative. In both stages, camera motion is specified explicitly by the trajectory condition rather than redundantly through language. This encourages the model to follow the provided trajectory while preserving the remaining prompt semantics.

3 Long-Horizon Video Generation

JoyAI-Echo-1.0 [Li et al., 2026] introduced a long-term audio-visual memory mechanism that reuses compact visual and acoustic cues from previous shots, substantially improving character appearance and speaker-timbre consistency during multi-shot generation. Building on this foundation, JoyAI-Echo-1.5 improves both long-range consistency and generation efficiency. First, it introduces a

unified audio-visual memory-conditioning framework that constructs visual memory from multiple shots, extracts audio memory from speech-filtered full-shot audio, separates memory and target tokens through RoPE coordinates, and jointly trains different combinations of memory and image conditions. Second, it applies distribution matching distillation to convert the original multi-step generator into an 8-step student, with optimization strategies designed for the distinct characteristics of video and audio generation. We first formulate the long-horizon generation task and present the unified memory-conditioning framework, and then describe the proposed distillation method for efficient audio-visual generation.

3.1 Unified Audio-Visual Memory Conditioning

Task formulation. Given a story script represented by shot-level text conditions $C = \{c_t\}_{t=1}^T$, JoyAI-Echo generates a sequence of synchronized audio-visual shots $S = \{S_t\}_{t=1}^T$, where $S_t = (V_t, A_t)$. Let M_0 denote an optional memory bank supplied before generation, and let $I = \{I_t\}_{t=1}^T$ denote optional image conditions for individual shots. Long-form generation is factorized as:

$$p_\theta(S \mid C, M_0, I) = \prod_{t=1}^T p_\theta(S_t \mid c_t, M_{t-1}, I_t), \quad M_t = \mathcal{U}(M_{t-1}, S_t). \quad (8)$$

Before generating shot t , M_{t-1} contains both the initial memory M_0 and the memory extracted from shots $S_1, \dots, S_{t-1}$. After S_t is generated, the update function $\mathcal{U}$ extracts its audio-visual information and updates the memory bank for the next shot. The image condition I_t , when provided, specifies the first frame of the current target shot and does not consume a memory slot.

Audio-visual memory construction. Each memory item is written as $m_i = (m_i^v, m_i^a)$, where m_i^v and m_i^a denote its visual and audio components. From the available history, we select up to K shots and arrange them in chronological order. For the i -th selected shot, one frame is sampled and encoded by the video VAE:

$$m_i^v = E_v(V_i[f_i]), \quad f_i \sim \mathcal{U}\{1, \dots, |V_i|\}. \quad (9)$$

The resulting latents provide compact visual observations from different historical shots instead of restricting the condition to a single adjacent frame.

For audio memory, we use the complete audio content of each selected shot. Since full-shot audio may contain background music, environmental sounds, and other non-speech components, we first apply a speech-filtering operator $\mathcal{F}_{\text{speech}}$ and then encode the filtered waveform using the audio VAE:

$$\tilde{A}_i = \mathcal{F}_{\text{speech}}(A_i), \quad m_i^a = E_a(\tilde{A}_i). \quad (10)$$

Here, $\mathcal{F}_{\text{speech}}$ is a generic speech-separation operator that can be instantiated by different filtering models; our current implementation uses a speech-extraction model through the Music-Source-Separation-Training (MSST) toolkit. In v1.0, a short audio window is selected according to acoustic energy. However, the maximum-response window may be dominated by music or other loud background components, which can be repeatedly reinforced when generated audio is reused during autoregressive rollout. The new construction instead uses the complete filtered speech content of each shot, providing cleaner and more stable speaker information. The visual and audio latents are then arranged in the same slot order, so that each memory item contains one compact visual observation and its corresponding event-level audio segment.

RoPE for memory conditioning. All historical conditions and user-provided memory inputs are represented in a common memory format. Following the discontinuous RoPE offsets introduced in ShotStream [Luo et al., 2026], we assign separated temporal RoPE regions to the current target, historical memory, and memory supplied before generation. The target video and audio tokens retain their original temporally aligned coordinates in a low-offset region below 500. When an image condition I_t is provided, its tokens remain in the coordinate system of the current target shot and are placed at temporal coordinate zero as its clean first frame.

The visual and audio components extracted from the i -th historical shot share the same temporal RoPE center:

$$\mu_i = 500 + 50(i - 1), \quad (11)$$

such that successive audio-visual memory items are centered at 500, 550, 600, $\dots$. All spatial tokens of the sampled visual memory frame use μ_i as their temporal coordinate while retaining their original

spatial RoPE coordinates. For the corresponding audio latent with native temporal coordinates $\{\tau_{i,\ell}^a\}_{\ell=1}^{L_i^a}$, we assign:

$$r_{i,\ell}^a = \mu_i + \tau_{i,\ell}^a - \bar{\tau}_i^a, \quad \bar{\tau}_i^a = \frac{1}{L_i^a} \sum_{\ell=1}^{L_i^a} \tau_{i,\ell}^a, \quad (12)$$

where ℓ indexes audio tokens. In other words, the temporal RoPE coordinates of the complete audio latent are shifted together so that its midpoint is aligned with the same center μ_i . This operation preserves the relative temporal structure of the audio event and does not modify the audio latent features themselves.

Memory supplied before generation is encoded independently and placed in a high-offset region:

$$M_0 = \{m_j^0\}_{j=1}^J, \quad m_j^0 = (m_j^{0,v}, m_j^{0,a}), \quad (13)$$

where

$$m_j^{0,v} = E_v(X_j^{\text{mem}}), \quad m_j^{0,a} = E_a(\mathcal{F}_{\text{speech}}(A_j^{\text{mem}})), \quad \rho_j \geq 5000. \quad (14)$$

The visual and audio components of the j -th initial memory item are both centered at ρ_j using the same positional assignment: the visual tokens use ρ_j as their temporal coordinate, while the audio RoPE coordinates are shifted together so that their midpoint is aligned with ρ_j . The centers $\{\rho_j\}_{j=1}^J$ are distinct and are assigned with enough spacing to keep the shifted audio-coordinate spans of different memory items non-overlapping. Consequently, the current target, historical memory, and initial memory occupy separated temporal RoPE regions. This separation keeps memory outside the target-shot timeline while allowing memory and image conditions to be provided simultaneously.

Memory interaction. After memory construction and RoPE assignment, memory and target tokens are jointly processed by the model. Unlike v1.0, v1.5 removes the memory-specific audio self-attention mask and the slot-paired cross-modal attention mask. All valid tokens interact through the original self-attention and bidirectional cross-modal attention paths. Their roles remain distinguishable because memory tokens use dedicated RoPE coordinates, remain clean, and are excluded from the prediction loss, whereas target tokens follow the target timeline and are optimized as generation outputs. This unified interaction removes additional hard constraints and naturally supports the joint use of memory and image conditions.

Joint conditional training. Training follows the two-stage data curriculum described in Sec. 2.1. Throughout the curriculum, we use a unified conditional interface that supports text-to-audio-video (T2AV), image-to-audio-video (I2AV), memory-conditioned text-to-audio-video (MT2AV), and memory-conditioned text-and-image-to-audio-video (MTI2AV). Let $b_M, b_I \in \{0, 1\}$ indicate whether audio-visual memory and a first-frame image condition are provided. The model condition for shot t is:

$$\mathcal{C}_t = (c_t, b_M M_{t-1}, b_I I_t), \quad (15)$$

where $(b_M, b_I) = (0, 0), (0, 1), (1, 0), (1, 1)$ correspond to T2AV, I2AV, MT2AV, and MTI2AV, respectively.

Let z_0^v and z_0^a denote the clean target video and audio latents, and let $\epsilon^v, \epsilon^a \sim \mathcal{N}(0, \mathbf{I})$. For each modality $q \in \{v, a\}$, rectified-flow training constructs the interpolation and target velocity for the tokens to be predicted:

$$z_\sigma^q = (1 - \sigma)z_0^q + \sigma\epsilon^q, \quad u^q = \epsilon^q - z_0^q. \quad (16)$$

We write the joint noised audio-visual target as $z_\sigma = (z_\sigma^v, z_\sigma^a)$ and the paired noise as $\epsilon = (\epsilon^v, \epsilon^a)$. Memory tokens are inserted as clean conditions with zero noise level and are excluded from the prediction loss. When $b_I = 1$, the first target video frame is also kept clean and excluded from the video loss. Let W_v and W_a be binary loss masks selecting only noised target tokens. The joint training objective is:

$$\mathcal{L} = \mathbb{E}_{\text{data}, \epsilon, \sigma, b_M, b_I} \left[\|W_v \odot (v_\theta^v(z_\sigma, \mathcal{C}_t) - u^v)\|_2^2 + \lambda_a \|W_a \odot (v_\theta^a(z_\sigma, \mathcal{C}_t) - u^a)\|_2^2 \right], \quad (17)$$

where λ_a balances the audio and video objectives. The resulting training distribution covers the four combinations of (b_M, b_I) , enabling generation without memory, with image conditioning alone, with memory alone, and with both conditions. This unified formulation supports T2AV, I2AV, MT2AV, and MTI2AV without changing the model interface or introducing separate objectives.

3.2 Acceleration with Distribution Matching Distillation

Long-form audio-visual generation requires both long-range consistency and low-latency inference. To accelerate JoyAI-Echo-1.5, we apply distribution matching distillation (DMD) to convert the original multi-step audio-visual generator into an 8-step student generator.

Data curation and training pipeline. JoyAI-Echo-1.5 performs DMD using memory conditions constructed from real long-form videos following Sec. 3.1. From the large-scale training corpus, we curate a compact subset with high-quality audio-visual memory and diverse generation conditions. The resulting data mixture covers T2AV, I2AV, and MT2AV, together with samples annotated by multi-shot captions. During training, we randomly sample these tasks according to predefined mixing ratios, allowing the student generator to preserve both single-shot generation quality and memory-conditioned multi-shot capability.

We find that DMD is particularly sensitive to the quality and distribution of visual memory. Low-level discrepancies in memory appearance, including brightness and saturation shifts, can bias the conditional distribution learned by the student and become progressively amplified when generated shots are written back into memory. Before DMD training, we therefore calibrate the low-level statistics of visual memory to better match the target training distribution. This preprocessing substantially improves distillation stability and reduces quality drift during multi-shot rollout.

The curated task mixture and calibrated memory conditions are used throughout the complete DMD pipeline. We first perform high-shift DMD with a relatively large learning rate to establish video quality and global structure, and then continue from the converged checkpoint with a lower shift and learning rate to refine audio and other fine-grained details. In both stages, the video branch is fully fine-tuned, while the audio and cross-modal modules are adapted through LoRA.

Asymmetric audio-visual optimization. Video and audio exhibit substantially different structures in the latent space. Video latents are highly structured and spatially redundant, with strong correlations across neighboring spatial locations and adjacent frames. Their temporal evolution is also relatively smooth, allowing global appearance, scene layout, and motion patterns to be represented through slowly varying latent structures. Audio latents, in contrast, encode dense one-dimensional temporal signals with rapidly changing local patterns. Fine-grained information such as phonetic content, pitch, transients, and acoustic textures may vary significantly within a short temporal interval [Su et al., 2026].

These differences lead to asymmetric optimization requirements during few-step DMD. The video branch must substantially adapt its spatial and temporal representations to recover visual structure and motion along the shortened sampling trajectory. Restricting its trainable capacity results in insufficient adaptation and degraded visual fidelity. The audio branch, however, is more sensitive to aggressive distribution-matching updates. Small deviations in the audio latent can remove perceptually important local details, while excessive parameter updates may overwrite the acoustic prior of the multi-step teacher and introduce muffled, metallic, or electronic-sounding artifacts [Tian et al., 2026].

We therefore allocate different trainable capacities to the two modalities:

$$\theta_G^{\text{train}} = \{ \theta_v^{\text{full}}, \Delta\theta_a^{\text{LoRA}}, \Delta\theta_{\text{cross}}^{\text{LoRA}} \}, \quad (18)$$

where the video branch is fully fine-tuned, while the audio branch and cross-modal attention modules are adapted through low-rank adaptation (LoRA). Full fine-tuning provides sufficient capacity for the video branch to learn the structural and motion changes required by few-step generation. In contrast, LoRA constrains the effective update space of the audio branch, slowing its deviation from the well-trained acoustic prior. Applying LoRA to the cross-modal attention modules preserves the flexibility required to adapt audio-visual correspondence without aggressively modifying the audio representation itself. This asymmetric parameterization therefore balances video adaptability with audio stability during joint DMD training.

Coarse-to-fine DMD noise scheduling. In DMD, the clean prediction produced by the student generator is re-noised to an auxiliary noise level σ , at which the teacher and fake-score model estimate the real and generated scores [Yin et al., 2024b,a]. The choice of σ is important because it determines the granularity of the distributional information exposed to score matching. Higher noise levels place greater emphasis on coarse distributional properties and help stabilize global structure, whereas lower

noise levels retain more sample-specific information and provide stronger supervision for fine-grained details.

Video and audio benefit differently from these noise regimes. Video generation relies strongly on high-noise DMD updates to establish stable appearance, scene layout, motion, and temporal structure under the shortened sampling trajectory. Audio quality, in contrast, is particularly sensitive to the low-noise regime, where local temporal patterns such as phonetic details, pitch, transients, and acoustic textures are more directly preserved and corrected. A fixed noise regime therefore cannot simultaneously provide the most suitable optimization signal for both modalities.

We address this mismatch with a two-stage DMD schedule. The first stage uses a relatively large timestep shift and learning rate, biasing score matching toward higher noise levels to establish stable video quality and structure. Starting from the converged first-stage checkpoint, the second stage reduces both the timestep shift and learning rate, shifting the optimization toward lower noise levels to refine audio quality and other fine-grained details. This coarse-to-fine schedule accommodates the different noise-level preferences of video and audio during joint few-step distillation.

Memory-robust and task-adaptive DMD. For the DMD stage described here, we consider T2AV and MT2AV, where MT2AV generates the current shot conditioned on the audio-visual memory bank M_{t-1} . Although the training memory is constructed from real long-form videos, the memory available during inference is progressively extracted from previously generated shots. Even small generation errors can therefore alter the memory distribution and accumulate over a multi-shot rollout. A student trained only with undegraded memory may become sensitive to this discrepancy, leading to gradual degradation of visual quality and cross-shot consistency.

To improve robustness, we apply stochastic degradation to the visual memory during DMD training. The student generator and fake-score model receive the same probabilistically degraded memory, allowing the fake-score model to track the student distribution under imperfect memory conditions. In contrast, the real-score model receives the original, undegraded memory and provides a stable target corresponding to the desired conditional distribution. The resulting score difference encourages the student to recover a high-quality current shot even when its historical memory contains moderate errors. This stochastic degradation complements the visual-memory distribution calibration described above: calibration corrects systematic low-level biases in the training memory, while degradation simulates the sample-dependent errors encountered during autoregressive rollout.

T2AV and MT2AV also exhibit different audio optimization dynamics. In T2AV, the student must construct the complete audio distribution from text and noise. In MT2AV, the memory condition already provides informative cues about speaker timbre, acoustic context, and temporal continuity, making its audio component easier to learn. Consequently, using the same audio weight for both tasks can cause the MT2AV audio objective to contribute disproportionately to joint DMD optimization.

We therefore introduce a task-dependent audio weight:

$$\mathcal{L}_{\text{DMD}} = \mathcal{L}_{\text{DMD}}^v + \lambda_a(\kappa)\mathcal{L}_{\text{DMD}}^a, \quad (19)$$

where $\kappa \in \{\text{T2AV}, \text{MT2AV}\}$ denotes the DMD task, with $\lambda_a(\text{T2AV}) = 0.25$ and $\lambda_a(\text{MT2AV}) = 0.10$. The reduced MT2AV weight prevents its easier memory-conditioned audio objective from dominating joint distillation, while the larger T2AV weight retains sufficient supervision for learning audio generation directly from noise. Together, stochastic memory degradation and task-adaptive weighting improve the robustness of few-step generation across both single-shot and memory-conditioned settings.

Audio distillation with adversarial and energy constraints. Despite the conservative Audio-LoRA parameterization and low-noise refinement stage, applying score-based DMD to audio remains challenging. The distribution-matching gradient mainly corrects the discrepancy between the teacher and student distributions at noisy latent states. For dense audio signals, this correction can favor dominant acoustic modes while suppressing subtle temporal variations, producing over-smoothed, muffled, or electronic-sounding outputs. We further observe that DMD may cause the overall audio latent level to increase continuously during training, leading to excessive loudness, clipping, and harsh distortion after decoding. We therefore introduce two complementary audio-specific objectives to constrain global signal level and local acoustic realism.

First, we match the root-mean-square (RMS) level of the generated audio latent to that of its ground-truth counterpart:

$$\mathcal{L}_{\text{energy}}^a = \psi(\log(\text{RMS}(\hat{z}_0^a) + \varepsilon_{\text{num}}) - \log(\text{RMS}(z_0^a) + \varepsilon_{\text{num}})), \quad (20)$$

where $\hat{z}_0^a$ and z_0^a denote the predicted and ground-truth clean audio latents, respectively. The robust penalty ψ limits the influence of extreme samples, and $\varepsilon_{\text{num}} > 0$ ensures numerical stability. Computing the discrepancy in the log domain makes the loss sensitive to relative rather than absolute level differences. This constraint suppresses persistent latent-level drift and reduces the risk of clipping after audio decoding.

Second, following AudioX-Turbo [Tian et al., 2026], we introduce a diffusion-based audio discriminator. We reuse intermediate features from the frozen audio teacher as the discriminator backbone and train only lightweight heads to distinguish generated and real audio latents. By preserving the pretrained audio representation and limiting adversarial training to the added heads, the discriminator provides a stable perceptual signal without modifying the teacher. Its local adversarial feedback complements the distribution-level DMD objective by preserving transients, acoustic textures, and other fine-grained temporal details.

The complete student-generator objective is

$$\mathcal{L}_G = \mathcal{L}_{\text{DMD}}^v + \lambda_a(\kappa)\mathcal{L}_{\text{DMD}}^a + \lambda_{\text{GAN}}\mathcal{L}_{\text{GAN}}^a + \lambda_{\text{energy}}\mathcal{L}_{\text{energy}}^a. \quad (21)$$

The DMD terms align the global audio-visual generation distribution, the adversarial loss restores local acoustic details, and the RMS-level constraint stabilizes the overall audio signal level. Together, these objectives mitigate both perceptual over-smoothing and excessive energy growth during few-step distillation.

4 Action-Conditioned World Modeling

4.1 Overview and Problem Formulation

Given an optional clean media context $\mathcal{M}$, a text condition $\mathcal{C}$, and a user control sequence $U_{1:T}$, our goal is to jointly generate synchronized video $V_{1:T}$ and audio $A_{1:T}$ under user-specified camera intent. The media context $\mathcal{M}$ accommodates an empty condition $\emptyset$, a single reference image I_{ref} , or an audio-visual prefix $(V_{1:\tau}, A_{1:\tau})$. We serialize user navigation inputs into a time-indexed metric camera trajectory event stream $\mathcal{E}_P = \{(t, P_t)\}_{t=1}^T$, where P_t represents a calibrated relative 6-DoF pose. The joint generation process is formulated as:

$$p_{\theta, \phi}(V_{1:T}, A_{1:T} \mid \mathcal{M}, \mathcal{C}, \mathcal{E}_P), \quad (22)$$

where θ denotes the audio-visual backbone, and ϕ represents the parallel relative trajectory-conditioning pathway. Static text fields specify scene semantics, while $\mathcal{E}_P$ dictates the continuous temporal trajectory evolution. When $\mathcal{M}$ contains an observed audio-visual prefix, those prefix samples are clamped and only the unobserved suffix is generated; the full-sequence notation in Eq. (22) is used for compactness.

4.2 Unified Camera-Intent Interface & Scale Calibration

Realizing continuous world modeling across heterogeneous datasets requires overcoming two key obstacles: the incompatibility of controller-specific action spaces and the scale mismatch of relative motion amplitudes. To address these issues, we decouple user-facing controls from model-level conditions through a two-fold strategy: (1) mapping arbitrary navigation inputs into a unified, subject-agnostic 6-DoF trajectory, and (2) calibrating relative translations with a robust global scale.

Relative 6-DoF Representation. We define camera intent using relative 6-DoF trajectories rather than actor-specific control states. Let $T_t \in \text{SE}(3)$ denote the camera-to-world extrinsic matrix at frame t . The relative motion between frames i and j is given by:

$$\Delta T_{ij} = T_i^{-1}T_j, \quad \xi_{ij} = \text{Log}(\Delta T_{ij})^\vee \in \mathbb{R}^6, \quad (23)$$

where ξ_{ij} is the corresponding 6-DoF twist coordinate. For discrete keyboard inputs $\mathbf{u}_t$, a fixed control mapping $\mathcal{F}_{\text{ctrl}}$ converts each input into a step-wise camera increment:

$$\xi_t = \mathcal{F}_{\text{ctrl}}(\mathbf{u}_t), \quad T_t = T_{t-1} \text{Exp}(\hat{\xi}_t), \quad \Delta T_t = T_0^{-1}T_t, \quad (24)$$

where $\hat{\xi}_t \in \mathfrak{se}(3)$. Continuous metric trajectories bypass $\mathcal{F}_{\text{ctrl}}$ and directly produce relative poses $\Delta T_{1:T}$. This unifies raw actions and estimated poses under a single geometric interface.

Global Translation Scale Calibration. To ensure consistent velocity semantics without introducing boundary artifacts in multi-turn rollouts, we calibrate relative translations across all data sources using a fixed global dataset-level scale s_{global} . For clip i , let $\Delta \mathbf{t}_{i,k}$ denote the relative translation at frame k with respect to the initial frame. We compute

$$s_{\text{global}} = Q_{0.9} \left(\left\{ \max_k \|\Delta \mathbf{t}_{i,k}\|_2 \right\}_{i \in \mathcal{D}_{\text{train}}} \right). \quad (25)$$

Relative translations are calibrated via $\hat{\mathbf{t}}_{i,k} = \Delta \mathbf{t}_{i,k} / s_{\text{global}}$, yielding $P_{i,k} = [\Delta \mathbf{R}_{i,k} \mid \hat{\mathbf{t}}_{i,k}]$ while keeping rotation matrices $\Delta \mathbf{R}_{i,k}$ intact.

4.3 Relative Trajectory Conditioning

Instead of absolute Plücker coordinates which depend on arbitrary world origins, we inject trajectory geometry through Unified Camera Positional Encoding (UCPE) [Zhang et al., 2025a]. UCPE makes pairwise camera geometry invariant to global coordinate shifts while preserving calibrated relative translation magnitudes.

For a latent token $i = (t, s)$ at frame t and spatial cell s , we define its local ray transformation $\mathbf{D}_i = (\mathbf{T}_i^{\text{wr}})^{-1}$ derived from calibrated pose P_t and intrinsics K_t . We inject this geometry into parallel attention heads via matrix scaling $\mathbf{G}_i = \mathbf{I}_{d/8} \otimes \mathbf{D}_i$ combined with spatiotemporal RoPE:

$$\begin{aligned} \tilde{\mathbf{q}}_i^c &= (\mathbf{G}_i^\top \oplus \text{RoPE}_i) \mathbf{q}_i^c, \\ (\tilde{\mathbf{k}}_i^c, \tilde{\mathbf{v}}_i^c) &= (\mathbf{G}_i^{-1} \oplus \text{RoPE}_i) (\mathbf{k}_i^c, \mathbf{v}_i^c), \\ \mathbf{o}_i^c &= (\mathbf{G}_i \oplus \text{RoPE}_i^{-1}) \text{Attn}_{\text{cam}}(\tilde{\mathbf{q}}^c, \tilde{\mathbf{k}}^c, \tilde{\mathbf{v}}^c)_i. \end{aligned} \quad (26)$$

Here, $\mathbf{T}_i^{\text{wr}}$ is the UCPE ray-to-world transform constructed from (P_t, K_t, s) , so $\mathbf{D}_i$ maps world coordinates to the ray-local frame; d is the attention-head width, and the superscript c identifies the parallel camera-conditioning stream. The symbol $\oplus$ denotes a block-diagonal direct sum with the spatiotemporal RoPE operator, and Attn_{cam} denotes attention in the camera pathway. The resulting camera-attention outputs pass through zero-initialized linear projections in the trajectory pathway parameterized by ϕ and are added to the main video self-attention blocks. The trajectory condition is applied exclusively to the visual backbone. One UCPE pathway is shared across first- and third-person data. Geometry alone does not determine whether a trajectory is realized from an egocentric or third-person observer–subject configuration, so the static viewpoint field supplies this initial relationship while $P_{1:T}$ supplies camera intent. This division allows the same trajectory representation and parameters to serve both viewpoints while leaving their camera–character realization to the learned world prior.

4.4 Progressive Training Curriculum

Data sources with rich audio-visual semantics and those providing precise, clean control trajectories are fundamentally heterogeneous. To resolve the resulting optimization conflicts, we employ a progressive three-stage curriculum that decouples audio-visual prior acquisition, control alignment, and joint consolidation.

Stage 1: Audio-Visual Continued Pretraining (AV-CPT). This stage adapts the base model to the visual styles, motion statistics, and acoustic dynamics (footsteps, ambient audio, and speech) of interactive game and general environments. We optimize the full backbone θ on the AV-rich data mixture while completely disabling trajectory conditioning. To simultaneously establish text-to-AV generation, image-to-AV generation, and temporal continuation capabilities, we randomly sample the clean media condition $\mathcal{M}$ from an empty set $\emptyset$, a single reference frame I_{ref} , or a synchronized audio-visual prefix $(V_{1:\tau}, A_{1:\tau})$. Unobserved video and audio latent regions are generated jointly, embedding strong acoustic priors and temporal continuity into the backbone.

Stage 2: Action Fine-Tuning (Action-SFT). This stage isolates control responsiveness without degrading the acquired audio-visual generation priors. We freeze the entire backbone θ and train only the lightweight UCPE camera pathway parameters ϕ on control-clean data using a visual-only denoising objective. To eliminate information shortcuts, we omit the dynamic-motion narrative and all audio-related fields, retaining only persistent scene, style, viewpoint, and subject descriptors $\mathcal{C}_{\text{static}}$. This forces the trajectory $\mathcal{E}_P$ to serve as the sole source of motion guidance. Furthermore, we heavily bias condition sampling toward reference-frame conditioning (I_{ref}), fixing initial scene content so the network concentrates parameter updates entirely on camera trajectory following.

Stage 3: Joint Fine-Tuning (Joint-FT). To consolidate control sensitivity with joint audio-visual synthesis, we unfreeze both the backbone θ and trajectory pathway ϕ for end-to-end co-adaptation on the balanced high-quality mixture. We restore environmental sound and speech fields to the text prompt (while keeping dynamic motion descriptions excluded) and resume joint audio-visual loss optimization. Training executes with a reduced learning rate to harmoniously align multimodal quality with trajectory responsiveness without collapsing the pre-trained priors.

5 Autoregressive Audio-Visual Generation

We progressively adapt a pretrained bidirectional, multi-step audio-visual diffusion model into a high-throughput, low-latency streaming autoregressive generator. This transition requires two key transformations: converting bidirectional temporal attention into causal autoregressive computation for online generation of synchronized audio-visual chunks, and compressing iterative denoising into a few-step sampler that remains stable under self-generated histories. To achieve this, we first apply audio-visual teacher forcing Williams and Zipser [1989], Chen et al. [2024] to initialize a causal streaming generator from the bidirectional backbone. We then introduce short-horizon Self-Gradient Forcing (SGF) Huang et al. [2026], Zhuang et al. [2026] to train on self-generated rollout states while distilling few-step generation, and further extend SGF to long-horizon self-rollouts to improve robustness and stability during sustained streaming inference.

5.1 Audio-Visual Teacher Forcing

We first use teacher forcing to initialize a causal generator from the pretrained bidirectional model. Specifically, during training, ground-truth audio-visual chunks are provided as the historical context, such that the generation of the current chunk depends only on preceding clean chunks rather than future observations. This stage provides a stable causal initialization before exposing the model to self-generated histories in subsequent self-forcing training.

Causal initialization. Let $m \in \{v, a\}$ index the modality, with v denoting video and a denoting audio. We divide both latent streams into N temporally aligned macro-chunks. The clean latent of modality m in chunk i is denoted by x_i^m , where $i \in \{1, \dots, N\}$. An aligned video-audio pair spans the same time interval, although the two chunks may contain different numbers of tokens. For each pair, we sample a shared noise level $\sigma_i \in (0, 1)$ and construct:

$$x_{\sigma_i, i}^m = (1 - \sigma_i)x_i^m + \sigma_i\epsilon_i^m, \quad \epsilon_i^m \sim \mathcal{N}(0, \mathbf{I}), \quad (27)$$

where ϵ_i^m is Gaussian noise and $\mathbf{I}$ is the identity covariance matrix. The shared σ_i places the paired audio and video chunks at a consistent diffusion stage.

To enable causal generation without sacrificing training parallelism, we concatenate the clean latents with their noisy counterparts. Let $\mathcal{B}_{i, \text{clean}}$ and $\mathcal{B}_{i, \text{noisy}}$ denote the clean and noisy latent blocks of the i -th paired audio-visual chunk, respectively. We impose the following causal attention pattern:

$$\mathcal{S}(\mathcal{B}_{i, \text{clean}}) = \{\mathcal{B}_{j, \text{clean}} : j \leq i\}, \quad \mathcal{S}(\mathcal{B}_{i, \text{noisy}}) = \{\mathcal{B}_{j, \text{clean}} : j < i\} \cup \{\mathcal{B}_{i, \text{noisy}}\}, \quad (28)$$

where $\mathcal{S}(\cdot)$ denotes the set of latent blocks accessible to the corresponding attention query. Under this mask, each noisy chunk can attend to all preceding clean audio-visual chunks and to the noisy tokens within the current chunk, while the corresponding clean target and all future chunks are masked out. We apply the same causal constraint to video self-attention, audio self-attention, and both directions of audio-visual cross-attention. For efficient causal attention computation, we implement this pattern using PyTorch FlexAttention [Dong et al., 2025] with a block-level causal mask. The resulting

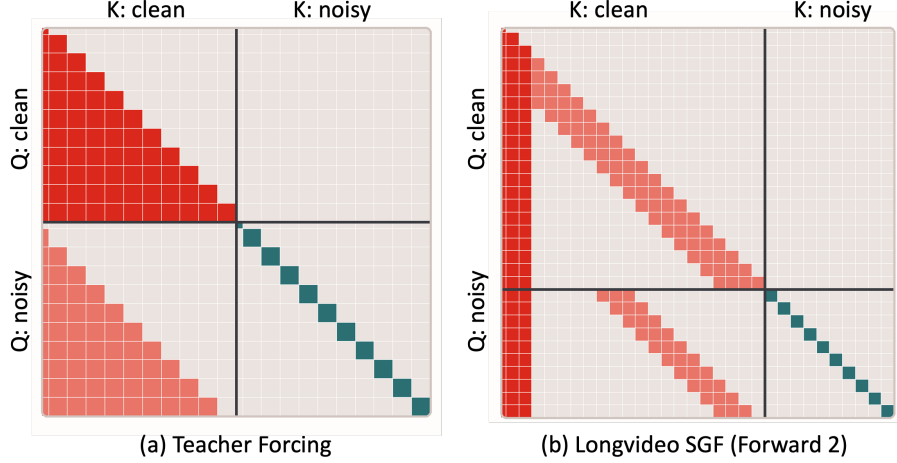


Figure 2: Illustration of the causal attention masks used in streaming post-training. (a) Audio-visual teacher forcing, where each noisy chunk attends to its preceding clean history and the current noisy chunk, while the clean target and future chunks are masked out. (b) Long-horizon SGF Forward 2, where the causal context is further restricted by the sink-plus-FIFO KV cache.

BlockMask exploits the structured sparsity of chunk-wise causal attention, allowing masked blocks to be skipped without materializing a dense attention mask. As illustrated in Fig. 2(a), this causal attention pattern enables each chunk to use the complete preceding clean history while preventing information leakage from the current clean target and future chunks. This converts the original bidirectional temporal attention into chunk-wise causal attention, while retaining parallel prediction of all chunks during training.

Training objective. To preserve the pretrained generative capability while efficiently adapting the model to the new causal attention pattern, we perform this stage using LoRA-based parameter-efficient fine-tuning Hu et al. [2021]. The pretrained backbone is kept fixed, while the LoRA parameters are optimized to learn the causal audio-visual generation behavior. Given the causally masked sequence, the model jointly predicts the flow velocities of the audio and video latents in a single forward pass. We retain the standard flow-matching objective of the pretrained diffusion model:

$$\mathcal{L}_{\text{TF}} = \sum_{m \in \{v, a\}} \lambda_m \mathbb{E} \left[\sum_{i=1}^N \|v_{\theta}^m(x_{\sigma_i, i}^m, \sigma_i) - (\epsilon_i^m - x_i^m)\|_2^2 \right], \quad (29)$$

where v_{θ}^m denotes the predicted flow velocity for modality m , λ_m balances the video and audio losses, and θ denotes the trainable LoRA parameters of the generator. Thus, audio-visual teacher forcing adapts the pretrained bidirectional diffusion model to causal, chunk-wise autoregressive generation while preserving its original flow-matching training objective.

5.2 Short-Horizon Self-Gradient Forcing

Starting from the causal teacher-forcing initialization, we further adopt Self-Gradient Forcing (SGF) [Zhuang et al., 2026], a checkpointed variant of Self Forcing Huang et al. [2026] that trains the generator on its own autoregressive histories to reduce the train–inference gap. Beyond standard Self Forcing, SGF reconstructs the self-generated context with gradient tracking, allowing losses from later chunks to supervise how preceding generated contexts are encoded into causal memory without backpropagating through the full sequential rollout. We further combine SGF with a DMD objective [Yin et al., 2024b] to distill the original multi-step diffusion model into a few-step generator, jointly improving autoregressive robustness and sampling efficiency.

Forward 1: Autoregressive self-rollout. The generator first performs a complete chunk-wise rollout with gradient computation disabled, using exactly the few-step sampler deployed at inference.

For example, with a four-step sampler, each audio-visual chunk is generated through four denoising steps, followed by an additional clean-context forward to prefill the causal KV cache for subsequent chunks. During the rollout, we sample an exit step t_e and record, for each chunk i , the corresponding noisy audio-visual latents $r_i = (r_i^v, r_i^a)$, together with the final generated clean latents $\hat{x}_i = (\hat{x}_i^v, \hat{x}_i^a)$. Since Forward 1 is executed under `no_grad`, it reproduces the inference-time autoregressive state distribution without retaining the computation graph of the full rollout.

Forward 2: Context-gradient reconstruction. We then reconstruct the rollout computation in a single differentiable forward using the states recorded in Forward 1. Specifically, the final generated clean latents $\{\hat{x}_i\}_{i=1}^N$ are evaluated at the clean context timestep $t = 0$, while the recorded noisy audio-visual latents $\{r_i\}_{i=1}^N$ retain their sampled exit timestep t_e . The two are concatenated following the same causal construction as Eq. (28), where each clean generated chunk serves as the prefetched causal context for subsequent chunks and each noisy latent corresponds to the current chunk at the sampled exit state. Applying the causal attention mask enables all chunks to be reconstructed in parallel, yielding the recovered clean predictions $\{\bar{x}_i\}_{i=1}^N$.

Although both the exit-step states and final generated latents are detached during Forward 1, their context representations are recomputed with gradient tracking in Forward 2. In particular, the $t = 0$ clean states reconstruct the KV-cache prefill computation through which generated chunks are encoded as causal history for subsequent generation. This allows losses on later chunks to propagate through the reconstructed K/V representations of preceding self-generated chunks, without backpropagating through the full sequential sampling trajectory.

DMD optimization. We optimize the generator with a DMD objective evaluated on the recovered predictions conditioned on self-generated autoregressive histories. Since these predictions are conditioned on the generator’s own autoregressive history rather than ground-truth context, the distillation objective simultaneously improves few-step sampling efficiency and mitigates error accumulation caused by self-generated histories.

The generator is initialized from the causal model obtained after the teacher-forcing stage, while both the real-score and fake-score models are initialized from the pretrained bidirectional diffusion model. The real-score model is kept frozen to represent the target data distribution, whereas the fake-score model is trainable and continuously updated to track the evolving distribution of the few-step generator.

For a sampled diffusion timestep s , we perturb the recovered clean audio-visual latents as:

$$\bar{x}_{s,i} = (1 - s)\bar{x}_i + s\eta_i, \quad \eta_i \sim \mathcal{N}(0, \mathbf{I}), \quad (30)$$

where $\bar{x}_i = (\bar{x}_i^v, \bar{x}_i^a)$ denotes the jointly recovered audio-visual prediction. The frozen real-score model s_{real} estimates the score of the pretrained data distribution, while the trainable fake-score model s_{fake} estimates the score of the current generator distribution. The generator is optimized with the DMD objective:

$$\mathcal{L}_{\text{SGF-DMD}} = \mathbb{E}_{i,s,\eta_i} [w(s) \langle \text{sg}(s_{\text{fake}}(\bar{x}_{s,i}, s) - s_{\text{real}}(\bar{x}_{s,i}, s)), \bar{x}_i \rangle], \quad (31)$$

where $\text{sg}(\cdot)$ denotes stop-gradient and $w(s)$ is the diffusion-dependent weighting factor. This objective drives the few-step generator toward the distribution represented by the pretrained multi-step model while preserving gradients through the reconstructed causal context.

Following each generator update, we perform multiple fake-score updates. The generated samples are detached and used to train the fake-score model with the standard flow-matching objective, allowing it to track the evolving generator distribution. This short-horizon SGF stage therefore adapts the generator to its deployment-time autoregressive states and establishes few-step generation before extending training to longer streaming horizons.

5.3 Long-Horizon Self-Gradient Forcing

After establishing stable few-step autoregressive generation with short-horizon SGF, we extend self-rollout to substantially longer synchronized audio-visual sequences. The training objective and few-step sampler remain unchanged, while the longer rollout exposes the generator to progressively accumulated prediction errors and the resulting shift in its autoregressive state distribution. This stage therefore explicitly adapts the generator to the error accumulation encountered during sustained streaming generation.

Long-horizon rollout construction. We construct a long-horizon trajectory by sequentially rolling out multiple temporal segments, each corresponding to the training horizon used in the short-horizon SGF stage. To maintain temporal continuity across segment boundaries, the terminal video frame of the k -th rollout segment is reused as the initial frame of the $(k + 1)$ -th segment. Consequently, adjacent rollout segments share a boundary frame, allowing generation to proceed continuously rather than restarting each segment from an independent initial state. Throughout the trajectory, all subsequent chunks are generated autoregressively from the model’s own preceding outputs.

Bounded causal context. A full causal KV cache grows with the rollout length and is therefore impractical for long-running streaming generation. We instead maintain a sink-plus-FIFO causal KV cache that preserves a small persistent prefix together with a bounded window of recent history. Let S_m denote the number of persistent sink chunks and R_m the number of recent chunks retained for modality m . The historical context retained in the KV cache when generating chunk i is:

$$\mathcal{H}_i^m = \underbrace{(\hat{x}_1^m, \dots, \hat{x}_{\min(S_m, i-1)}^m)}_{\text{attention sink}} \parallel \underbrace{(\hat{x}_{\max(S_m+1, i-R_m)}^m, \dots, \hat{x}_{i-1}^m)}_{\text{recent FIFO history}}, \quad (32)$$

where $\parallel$ denotes ordered concatenation, and an indexed range is empty when its lower bound exceeds its upper bound. Thus, $\mathcal{H}_i^m$ contains only previously generated chunks: the attention sink preserves a fixed long-term prefix, while the FIFO cache maintains the most recent context as generation proceeds. Accordingly, the retained history is bounded by $|\mathcal{H}_i^m| \leq S_m + R_m$, making the KV-cache size independent of the total rollout duration. The same sink-plus-FIFO policy is used during training and inference.

As illustrated in Fig. 2(b), Forward 2 applies this sink-plus-FIFO pattern to the reconstructed clean context and the current noisy chunk. Importantly, the generator and the DMD score models use different positional views of this long-horizon context. The generator follows the streaming causal cache described above, whereas both the real-score and fake-score models are bidirectional models inherited from the pretrained fixed-length model and should therefore only be evaluated within the temporal range observed during pretraining. For each DMD evaluation, we restrict the score-model conditioning context to a fixed-size temporal window and rebase its temporal positional indices to the canonical pretrained range. In effect, the retained sink chunks, FIFO history, and current chunk are always mapped into the same positional range as the first pretrained temporal window, preventing the score models from operating on out-of-distribution temporal positions during long-horizon rollout.

Long-horizon DMD supervision. We apply the DMD objective to every rollout segment rather than supervising only the final segment of the long trajectory. Let $\mathcal{L}_{\text{DMD}}^{(k)}$ denote the DMD loss computed from the recovered predictions of the k -th rollout segment. For a trajectory containing K segments, the long-horizon objective is accumulated across all segments:

$$\mathcal{L}_{\text{Long-SGF}} = \sum_{k=1}^K \mathcal{L}_{\text{DMD}}^{(k)}. \quad (33)$$

Thus, supervision is applied throughout the entire self-generated trajectory: the loss from later rollout segments is evaluated under histories that already contain errors accumulated in earlier segments. Jointly optimizing these segment-wise losses encourages the few-step generator to remain stable across progressively shifted autoregressive states, rather than adapting only to a single short-horizon rollout distribution.

5.4 Autoregressive Inference

At inference, the generator produces synchronized audio-visual chunks autoregressively using the same few-step sampler and sink-plus-FIFO cache introduced during training. For each new chunk, the retained modality-specific histories $\mathcal{H}_i = (\mathcal{H}_i^v, \mathcal{H}_i^a)$ provide the causal context, while outdated FIFO entries are discarded to keep the KV cache bounded. Consistent with long-horizon training, the temporal RoPE indices of the sink, FIFO, and current chunks are rebased into a fixed positional window as generation proceeds, maintaining a consistent temporal positional layout throughout streaming inference.

Let Θ collect the active generator parameters (including the trajectory pathway ϕ for the world-model variant), and let $\mathcal{D}_i$ denote the variant-specific conditions: text together with memory and/or an image

for the long-video variant, or media context, text, and trajectory for the world-model variant. The next synchronized chunk is sampled as:

$$(V_i, A_i) \sim p_{\Theta}(\cdot \mid \mathcal{H}_i, \mathcal{D}_i), \quad (34)$$

where V_i and A_i denote the generated video and audio chunks, respectively.

5.5 Towards Causal Memory

Teacher forcing in Eq. (28), long-horizon SGF, and inference already share one causal rule. A noisy chunk may attend only to previously generated clean history. The current clean target and all future chunks remain masked. The sink-plus-FIFO cache in Eq. (32) is the bounded form of this rule. The attention sink preserves a fixed early prefix, the FIFO retains the most recent chunks, and temporal RoPE within each $\mathcal{H}_i^m$ is rebased as outdated FIFO entries are discarded, so that $\mathcal{H}_i = (\mathcal{H}_i^v, \mathcal{H}_i^a)$ keeps a consistent positional layout throughout streaming generation. This cache is sufficient for local audio-visual continuity under the few-step sampler. It is not sufficient once a chunk has left the FIFO. Those tokens are no longer available, even if the sink still holds the opening of the stream. When the camera returns, or when a model trajectory revisits a previously observed place, the generator should not treat the scene as new merely because the corresponding history has been evicted.

To retain structure beyond the sliding window without changing the causal mask, the few-step sampler, or the variant-specific conditions $\mathcal{D}_i$ in Eq. (34), we introduce a compact recurrent state $\mathcal{M}_i$ that is written only from the past. Let C_i denote a short leak-free prefix taken strictly before the start of chunk i , and let $\psi \subset \Theta$ collect the memory parameters inside the generator, including the trajectory pathway ϕ when the world-model variant is used. The next synchronized chunk and the updated memory are

$$(V_i, A_i) \sim p_{\Theta}(\cdot \mid \mathcal{H}_i, \mathcal{M}_{i-1}, \mathcal{D}_i), \quad \mathcal{M}_i = \mathcal{F}_{\psi}(\mathcal{M}_{i-1}, C_i, V_i). \quad (35)$$

Thus $\mathcal{H}_i$ continues to bound attention, while $\mathcal{M}_i$ stores what should remain after the FIFO forgets.

Leak-free prefix. The construction of C_i follows the same constraint as Eq. (28). Every index in C_i satisfies $t < t_i^{\text{start}}$, so the write never observes the current clean target or any future chunk. The prefix is ordered from oldest to newest and is concatenated in front of the noisy target, so a left-to-right scan matches the chunk order of streaming generation. First-chunk windows that do not contain such a history are dropped during training. Retaining them forces a first-frame fallback and would allow $\mathcal{F}_{\psi}$ to condition on a target that is still being denoised, which is the analogue of letting $\mathcal{B}_{i,\text{noisy}}$ attend to $\mathcal{B}_{i,\text{clean}}$.

Target-first rebase. As the FIFO slides, we already rebase temporal RoPE on $\mathcal{H}_i^m$. Camera and trajectory encodings that travel with C_i are rebased in the same way, using the first pose of the current chunk as the origin rather than the origin of chunk $i - 1$. Without this rebase, $\mathcal{M}_{i-1}$ and C_i would remain in the previous chunk’s coordinate system, and the trajectory pathway ϕ would observe an artificial jump at every chunk boundary even when the underlying trajectory is smooth.

Block-wise recurrent write. $\mathcal{F}_{\psi}$ is realized as a sparse state-space scan attached to selected DiT blocks, rather than as a global memory token in text cross-attention. For each spatial token trajectory the module maintains a state s_t and updates

$$s_t = \sigma(\delta) \odot s_{t-1} + (1 - \sigma(\delta)) \odot \tilde{u}_t, \quad y_t = g_t \odot s_t, \quad (36)$$

where δ is a learned per-channel decay, $\tilde{u}_t$ is a bounded input projection of the normalized block features, and g_t is a sigmoid write gate. The scan runs over the concatenated prefix and target, so the residual at the start of chunk i already contains a summary of C_i . The scan output is added through a residual whose scale is confined to a small interval. The lower bound prevents the recurrence from collapsing to a no-op, and the upper bound prevents it from overwriting the pretrained backbone. After V_i is generated, we retain only the last few latent-aligned steps as C_{i+1} and take the terminal scan state as $\mathcal{M}_i$. Long-video variants continue to place text and optional image conditions in $\mathcal{D}_i$, and world-model variants continue to condition on ϕ . Causal memory does not replace those inputs. It is the component of the history that should remain available after $\mathcal{H}_i$ slides.

The three components admit natural alternatives. C_i can be replaced by a suffix, a nearest-first tail, or FoV retrieval. Actions can keep the previous chunk’s coordinates. $\mathcal{F}_{\psi}$ can be omitted, given

an unbounded residual, or trained on first-chunk windows that have no real prefix. We compared these variants under the same backbone, chunk length, and SGF schedule. The configuration above, leak-free prefix, target-first rebase, bounded block-wise write, and history-bearing training windows, is the one that worked best. It does not change Eq. (34). It only specifies what $\mathcal{H}_i$ contains and how much of that history is written into $\mathcal{M}_i$ before the sink-plus-FIFO window slides.

6 Agentic Planning and Control

Long-form video creation begins with an open-ended user intention, whereas a video generator requires concrete descriptions of what happens in each shot. A short request rarely specifies a complete narrative, stable character identities, scene transitions, dialogue, camera progression, or which earlier visual evidence should be reused. We therefore introduce a Director Agent that organizes generation as a persistent, shot-wise process. The agent develops the user’s intention into a screenplay, applies prompt enhancement (PE), manages visual–audio memory, and coordinates generation and local revision. The video model remains responsible for synthesis, while the agent determines what should be generated and which context should condition it.

A central goal is autonomous completion. After receiving a brief and any optional reference media, the agent can understand the request, plan the story, prepare every shot, generate the synchronized long audio-video sequence, check the intermediate results, update continuity memory, and assemble the final film without waiting for a human decision at every shot. This is the default autonomous path. The same path exposes human-in-the-loop checkpoints: a user can act as a critic, edit the screenplay or current prompt in real time, reselect a memory, and resume generation from the affected shot.

The workflow is organized around a persistent project workspace. The workspace contains the user request, the evolving screenplay, a compact story profile, the ordered shot plan, generated artifacts, the character and scene memory banks, and the decisions made during review. It also records the relation between neighboring shots and the references supplied to each generation request. This state is important for long videos: a later shot is conditioned on accepted visual evidence and on the user’s latest decisions, rather than on a reconstruction of the conversation or on an unverified plan. The workspace can therefore be resumed from an intermediate shot, while a revision changes only the affected shot and its downstream state.

Recent agentic workflows have been explored for long-video generation [HyCreator Team, 2026]. Our workflow separates three responsibilities that are easy to conflate in a long-video pipeline. Planning decides what the story should contain and how it is divided into observable beats. Generation turns one prepared beat into synchronized audio and video under the selected references. Review and memory admission decide which result is accepted and which visual evidence is safe to expose to later shots. Keeping these decisions explicit makes the workflow inspectable: a failed generation can be revised without silently changing the screenplay, and an unsuitable memory proposal can be replaced without altering the accepted shot.

6.1 Planning and Prompt Enhancement

Read the brief and develop the screenplay. Given a user instruction u , the agent first resolves the central event, characters, locations, temporal progression, and intended ending. It produces a complete screenplay before the first generation call, but keeps the individual shot descriptions editable. The screenplay establishes the narrative order, one main dramatic job for each outer shot, and the relation between adjacent shots. An outer shot is the unit submitted to the video generator; any internal camera changes needed to expose one beat remain part of that shot. In the H3 deployment, the outer duration is bounded by ten seconds, so the planner places the main event early enough to finish and holds the final state long enough for it to be visually and audibly verified.

The planning record contains a compact story profile in addition to prose. It gives every recurring subject a stable identity, appearance, wardrobe, and voice anchor, and records the scene, important props, viewpoint, and relevant sound context. Stable subject IDs are reused across all later captions. This allows a later mention of a character to refer to an established identity while still permitting the shot to specify a new pose, action, or camera relationship. The profile also distinguishes persistent facts from shot-local facts. For example, a character’s coat and voice can persist, whereas a wet sleeve or a temporary facial reaction belongs only to the shot in which it is observed.

Prepare the shot plan. For every adjacent pair, the planner marks one of three continuity relations: a scene transition, an ordinary edit within the same scene, or one continuous action. A scene transition changes the location or the established spatial state and starts with fresh scene context. An ordinary edit keeps the scene but changes the viewpoint or shot composition. A continuous action preserves the terminal state and motion direction of the previous shot. This relation is written into the shot plan before generation and is later used consistently when choosing a representative scene frame, a true tail frame, and the image references described in the caption.

Construct the production prompt. The screenplay and the PE captions are generated as one planning stage:

$$C = \{c_t\}_{t=1}^T = \text{PE}(\mathcal{D}(u)). \quad (37)$$

Here $\mathcal{D}$ produces the narrative plan, c_t is the enhanced caption for outer shot t , and T is the number of planned shots. We use $S_t = (V_t, A_t)$ for the synchronized shot, following Sec. 3.1. The equation describes the interface between planning and generation rather than a new learning objective: the agent writes the conditions and the video model samples the corresponding shot.

Structured caption fields. PE treats a caption as a set of coordinated fields rather than as an undifferentiated paragraph. For shot t , we write the organization abstractly as

$$c_t = [r_t^{\text{story}}; r_t^{\text{visual}}; r_t^{\text{camera}}; r_t^{\text{audio}}; r_t^{\text{temporal}}], \quad (38)$$

where the five terms describe the narrative beat, visible subjects and environment, camera and composition, synchronized sound, and the ordered changes within the shot. The fields are written as natural language in the final prompt; the notation only makes their roles explicit.

The story field contains the subjects, action, dialogue, and required ending. The visual field records appearance, clothing, props, scene geometry, lighting, and style. The camera field specifies shot scale, viewpoint, composition, and motivated movement, while the audio field aligns dialogue, Foley, ambience, and music with the visual event. The temporal field gives the initial state, trigger, preparation, continuous action, completion, reaction, and final hold. The fields preserve our identity anchors, scene relations, and audio–visual memory roles while keeping the production plan readable at each stage.

Executable detail and grounding. PE preserves the hard invariants of the request—subjects, identities, props, dialogue, language, duration, and outcome—and expands each beat as initial state $\rightarrow$ trigger $\rightarrow$ action $\rightarrow$ completion $\rightarrow$ reaction $\rightarrow$ final hold. It adds only observable evidence that improves execution, such as contact mechanics, environment response, or material-specific sound; it does not introduce a new event, subject, or outcome merely to make the caption longer.

Camera, sound, and references. The camera field is chosen to expose the evidence: an establishing view gives geography, a closer view gives expression or manipulation, and a motivated reframe reveals a consequence. Dialogue, Foley, ambience, and music follow the same timeline. Identity references provide appearance and voice cues, scene references provide layout and lighting, and an opening tail provides the starting state for continuous action. PE names only references supplied to the current shot, keeping text and media conditions consistent.

Initial and image-aware PE. An initial reference image grounds the screenplay in its visible people, objects, composition, viewpoint, and lighting. Immediately before an image-conditioned generation, a second PE pass describes how that state evolves into the planned beat. It preserves subject IDs, dialogue, language, and reference roles while adapting the opening pose and motion. The same rule is used for a user-provided first frame and for the tail frame of an accepted shot. The user may revise the screenplay and shot count before generation; afterward, the confirmed anchors constrain local prompt edits.

6.2 Short-Video Generation

Short-video generation uses the same planning, PE, optional image conditioning, generation, and user revision procedure as the long-video workflow, but treats the request as a single-shot screenplay ($T = 1$). The agent still resolves the event, starting state, ending state, camera observation, and

synchronized sound; the difference is that no later shot needs to inherit the result. For a fresh text-to-audio-video request, the enhanced caption is sent directly to the generator. The generated audio and video are sampled together so that the action, speech, Foley, and ambience follow one timeline.

With a user-provided first-frame reference, a multimodal model interprets the visible pose, viewpoint, environment, lighting, and terminal action state. It performs a local image-aware rewrite of the enhanced caption: the opening pose and state-dependent action are adjusted to follow naturally from the frame, while the story beat, subject identities, dialogue, language, and reference roles remain fixed. The first frame is then passed as the image condition. A fresh shot has no image condition; an image-conditioned shot begins from the provided visual state. The same interface supports the audio-visual model’s text, image, and memory conditioning without introducing a separate short-video prompt format.

In autonomous mode, the agent completes the short-video path using its default review and delivery policy. In human-in-the-loop mode, the rendered clip is shown to the user as a critic checkpoint. The user can provide a concrete correction—for example, an appearance, action, dialogue, camera, sound, or continuity change—and the agent rewrites the current caption before regenerating it. Since there is no following shot, the accepted short clip does not update a cross-shot character bank. This keeps the single-shot path simple while preserving the same real-time revision semantics used by the long-video path.

6.3 Shot-Wise Long-Video Generation

Autonomous execution. The agent executes the confirmed screenplay in timeline order and prepares the next shot only after the current shot has been resolved. Auto Mode runs this loop from the brief to the final merge with default review and memory policies. The interactive mode exposes the same states to a human critic, who can pause the current shot, modify its prompt or memory choice, and resume from there.

Memory and shot condition. The character memory bank is partitioned by stable character ID, $B_{t-1} = \{B_{t-1}^{(k)}\}_k$, and stores an approved visual observation, paired audio evidence, and its source shot. A separate full-frame scene record preserves same-scene layout; a true tail frame is reserved for continuous action. Scene transitions start with fresh scene context.

Before generating shot t , the agent retrieves the entries for the identities in c_t . Let $m_{t-1}^{(k)}$ denote the selected entry for character k , and let P_t denote the same-scene representative; $P_t = \emptyset$ after a scene transition. The active memory is

$$M_{t-1} = \left\{ m_{t-1}^{(k)} \mid k \in \text{IDs}(c_t) \right\} \cup P_t. \quad (39)$$

Here $\text{IDs}(c_t)$ lists the current subjects. Character entries provide identity and speaker cues, while P_t provides scene context. If the user requests another existing memory, the agent rebuilds M_{t-1} and regenerates only the current shot.

Image-aware shot preparation. The agent supports fresh generation, first-frame generation, same-scene reference generation, and continuous generation. Let I_t be the optional image condition: a user first frame, the accepted previous tail, or $\emptyset$ for a fresh shot. Image-aware PE rewrites only the opening state needed to connect c_t to I_t and preserves the story beat, subject IDs, dialogue, and reference roles. The shot is sampled as

$$S_t \sim p_\theta(\cdot \mid \tilde{c}_t, M_{t-1}, I_t), \quad S_t = (V_t, A_t), \quad (40)$$

where $\tilde{c}_t$ is the image-aware caption, M_{t-1} is active memory, and I_t fixes the starting visual state.

Layered review and real-time intervention. After generation, Auto Mode continues with its default policy. In the interactive path, the user can accept the clip or provide critic feedback on appearance, action, dialogue, camera, sound, or continuity. The agent edits the current caption or active memory and regenerates only that shot; rejected artifacts do not enter later context.

Memory admission after shot acceptance. After an accepted shot, candidate frames and synchronized audio are collected. A VLM proposes a subject crop and a scene representative; Auto Mode resolves the proposal by default, while the user may approve or reselect it. Character crops enter the corresponding bank, the scene frame remains continuity-only, and the true tail is kept for continuous generation.

Let $\hat{m}_t^{(k)}$ denote the resolved character memory for identity k after the automatic policy or the user’s review. The bank update can then be written as

$$B_t^{(k)} = \begin{cases} \text{Update}\left(B_{t-1}^{(k)}, \hat{m}_t^{(k)}\right), & \hat{m}_t^{(k)} \text{ is admitted,} \\ B_{t-1}^{(k)}, & \text{otherwise,} \end{cases} \quad (41)$$

where admission is decided automatically or by the user after re-selection. An unresolved proposal leaves the previous bank unchanged.

Assemble and deliver. After memory resolution, the agent prepares the next shot and repeats the loop until all planned shots are accepted. It then merges them in screenplay order with synchronized audio. Auto Mode delivers the completed film directly; interactive mode permits a final whole-film review before delivery.

7 Super-Resolution

The generators of the preceding sections produce synchronized audio-visual content at the 720-class resolutions of the training corpus (Sec. 2.1.2); a dedicated super-resolution (SR) stage then restores fine texture and suppresses generation artifacts at the delivery resolution. This factorization concentrates the dominant inference cost in the SR stage: a diffusion-based SR model must evaluate a large backbone at the full output resolution, and it must do so for every frame of the content it refines. Because resolution is fixed by the deliverable and duration by the content, the per-evaluation cost of the SR network is not negotiable; the number of network evaluations is the only lever that can be moved without changing what is delivered. Existing few-step distillation methods pull this lever at training time, baking a single step count into the weights [Yin et al., 2024b,a, ?], while training-free ODE solvers [??] preserve flexibility but degrade sharply in the very-few-step regime that matters most for high-resolution video. We take a third route: we train a flow-matching SR teacher [?] on the Wan2.1 backbone [Wan Team et al., 2025] and distill it with MeanFlow [?] into a model of the *average* velocity over an interval, for which the step count becomes an inference-time dial rather than a retraining decision. One set of weights supports 1-, 2-, 4-, or generally N -step sampling with no retraining, no auxiliary networks, and no external solver schedule. Sec. 7.1 describes the teacher, Sec. 7.2 the distillation and the resulting sampler, and Sec. 7.3 the lossless warm start that connects the two.

7.1 SR as Latent Restoration

Conditioning by latent concatenation. We cast video SR as conditional restoration in the latent space of the Wan2.1 VAE [Wan Team et al., 2025], which compresses video by $4\times$ temporally and $8\times$ spatially into 16 latent channels. The low-quality (LQ) input is first resized in pixel space to the target resolution by inexpensive bilinear interpolation, so that magnification is delegated to a fixed resampler and the network solves a pure restoration problem at a single scale: the condition and the target occupy the same latent grid, and the model’s task is to re-synthesize the high-frequency content that resizing cannot recover. The resized LQ video is encoded by the VAE, its latent statistics are aligned to the generated-video domain by a per-channel affine map estimated offline from paired degraded and model-generated samples, and the resulting 16-channel condition latent z^{LQ} is concatenated with a 4-channel all-ones mask $\mathbf{1}$ to form a 20-channel condition. The backbone is a DiT with the Wan2.1-T2V-1.3B geometry (30 blocks, hidden dimension 1536, umT5 text conditioning) whose patch embedding is widened from 16 to 36 input channels, so that the noisy target latent and the condition enter the network jointly at the first layer. To discourage the trivial shortcut of copying the condition, the full 20-channel condition is stochastically corrupted,

$$y = (1 - \alpha) [\mathbf{1}; z^{\text{LQ}}] + \alpha \epsilon_y, \quad \alpha \sim \mathcal{U}(0.3, 0.6), \quad \epsilon_y \sim \mathcal{N}(0, \mathbf{I}), \quad (42)$$

with α redrawn on every forward pass; the same perturbation is applied at training and at sampling time.

Degradation-aware data construction. The teacher is trained on paired real high-quality/low-quality videos, with a synthetic degradation pipeline layered on top of the already-degraded LQ side so that the condition distribution covers both real transmission artifacts and the failure modes of generated video. The synthetic pipeline composes three stages: (i) temporally coherent video degradations whose parameters follow low-frequency trajectories over time, including ghosting via temporal morphing of adjacent frames, stochastic frame dropping with interpolated reconstruction, directional motion blur with smoothly varying kernels, and low-frequency spatial warping localized to a drifting region of interest; (ii) a genuine codec compression round-trip; and (iii) Real-ESRGAN-style image-space degradations [?] (blur, resampling, noise, and JPEG compression), following real-world video SR practice [?], terminated by a down-up resampling step that destroys high-frequency information while preserving the target size. The strength of the synthetic degradations was staged over the course of teacher training, ending with a comparatively weak setting. On the target side, an in-batch perceptual-quality gate retains only high-quality clips and replaces rejected samples with temporally flipped copies of retained ones, keeping the supervision signal clean without shrinking the effective batch.

Teacher objective. The teacher is trained with the standard conditional flow-matching objective [?], consistent with the convention of Eq. (27): for a clean latent x_0 , noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, and noise level $t \in (0, 1)$, the noisy latent is $x_t = (1 - t)x_0 + t\epsilon$ and the network regresses the straight-path velocity,

$$\mathcal{L}_{\text{SR}} = \mathbb{E}_{x_0, y, t, \epsilon} \left[\|v_\theta(x_t, t, y) - (\epsilon - x_0)\|_2^2 \right], \quad (43)$$

where t is drawn from a logit-normal distribution warped by a time-shift transform that concentrates training density at higher noise levels, y is the noised condition of Eq. (42), and text conditioning is provided by the caption. The teacher is obtained by approximately 10.4k optimizer steps of SR fine-tuning through a staged curriculum over degradation strength and resolution, ending on multi-aspect-ratio buckets that match the intended delivery resolutions. At inference, this teacher is a conventional multi-step flow model: producing a single output requires tens of solver steps [?], each a full forward pass at the output resolution, which is precisely the cost that the next subsection removes.

7.2 Distilling an Arbitrary-Step Sampler

Average velocity. MeanFlow [?] replaces the instantaneous velocity $v(x_t, t)$ with the average velocity over an interval $[r, t]$,

$$u(x_t, r, t) = \frac{1}{t - r} \int_r^t v(x_\tau, \tau) d\tau, \quad 0 \leq r < t \leq 1, \quad (44)$$

so that the exact solution of the probability-flow ODE over $[r, t]$ is the single subtraction $x_r = x_t - (t - r)u(x_t, r, t)$. A network $u_\theta(x_t, r, t)$ that models this quantity is an interval integrator by construction: it does not approximate a step of an ODE solver, it *is* the step, for any interval it is queried on.

MeanFlow identity and training objective. Differentiating Eq. (44) along the trajectory yields the MeanFlow identity

$$u(x_t, r, t) = v(x_t, t) - (t - r) \frac{d}{dt} u(x_t, r, t), \quad \frac{du}{dt} = \partial_x u \cdot v + \partial_t u, \quad (45)$$

which converts the intractable interval integral into a local regression target: the known conditional velocity $\epsilon - x_0$ corrected by a Jacobian-vector product (JVP) of the network itself, evaluated under stop-gradient. We train with an explicit-gradient form of this regression,

$$\mathcal{L}_{\text{MF}} = \mathbb{E} \left[\|u_\theta - \text{sg}(u_\theta + w \Delta)\|_2^2 \right], \quad \Delta = (\epsilon - x_0) - u_\theta - \lambda(t - r) \widehat{\frac{du}{dt}}, \quad (46)$$

where $\text{sg}(\cdot)$ denotes stop-gradient and $\widehat{\frac{du}{dt}}$ is a stop-gradient estimate of the total derivative, so that the gradient with respect to u_θ is $-2w\Delta$ and the update direction is set directly. The per-sample weight $w = 1/(\|\Delta\| + c)$ is computed under stop-gradient—a gradient-carrying weight would open a backpropagation path that rewards enlarging the loss to shrink the weight—and down-weights

high-noise samples where the JVP estimate is least stable. The tangent norm entering w is rescaled to be invariant to spatial and temporal tensor shape, which keeps a single c valid across all resolution buckets. The factor λ linearly ramps from 0 to 1 over the first 2,000 steps, interpolating smoothly from pure flow matching to the full MeanFlow objective while the network acquires its interval dependence. The pair (t, r) is drawn as the maximum and minimum of two independent samples from the same shifted logit-normal family used for teacher training (time-shift 2.0 in the distillation stage), so no new timestep schedule is introduced at distillation time and every sample trains the genuine interval objective ($t > r$).

Jacobian–vector products by finite differences. Exact forward-mode JVPs are unavailable in our training stack: fused attention kernels do not implement forward-mode rules, non-reentrant activation checkpointing does not propagate tangents through the recomputed forward, and the sharded data-parallel backward does not support them. We therefore estimate du/dt by a central finite difference *along the ODE trajectory*: perturbing (x_t, t) jointly by $\pm h$ in the direction $(v, 1)$ captures both the $\partial_x u \cdot v$ and $\partial_t u$ terms of Eq. (45) in a single difference. The two probe evaluations run without gradient tracking, with network outputs kept in full precision and the differencing performed in double precision, since half-precision mantissas are destroyed by the near-cancelling subtraction; both probes share an identical random state and a pinned condition tensor y , so that the stochastic conditioning noise of Eq. (42) cannot leak into the difference. One-sided differences are substituted per sample when the interval boundary would be crossed. The step size h was calibrated by sweeping against a higher-precision reference evaluation of the production model—an internal engineering measurement at a single operating point rather than an ablation—and the chosen value sits in the regime where truncation and round-off errors balance; the adaptive weight and warmup in Eq. (46) absorb the residual magnitude error, as the objective chiefly consumes the tangent’s direction. Measured on our configuration, the estimator costs $2.0\times$ the step time of plain flow matching and no additional activation memory, since peak memory remains set by the single gradient-carrying forward.

Arbitrary-step sampling. The distilled model needs no scheduler, no solver, and no per-budget retraining. Given any budget of K network evaluations, we partition the noise axis $1 = t_0 > t_1 > \dots > t_K = 0$ by warping a uniform grid through the same time-shift transform used in training,

$$t_k = \frac{s(1 - \frac{k}{K})}{1 + (s-1)(1 - \frac{k}{K})}, \quad k = 0, \dots, K, \quad (47)$$

with shift $s = 2.0$ and the endpoints fixed to the maximal training noise level and to zero, which concentrates the few available evaluations in the high-noise region where training density was placed, and iterate the exact interval update

$$x_{t_{k+1}} = x_{t_k} - (t_k - t_{k+1}) u_\theta(x_{t_k}, t_{k+1}, t_k), \quad k = 0, \dots, K-1. \quad (48)$$

The sampler consists solely of the grid of Eq. (47) and the subtraction in Eq. (48); no scheduler object exists. Setting $K = 1$ gives one-step generation; larger K trades computation for fidelity along a continuum, and during training we render $K \in \{1, 2, 4\}$ side by side from the same checkpoint at every validation. This is the deployment property we set out to obtain: the SR cost per output pixel becomes K forward passes with K chosen at request time, so a single model serves latency-critical interactive use at one step and quality-leaning offline rendering at higher step counts, replacing the tens of solver steps required by the teacher. We emphasize what this is not. It is not a fixed-step distilled student in the sense of Secs. 3.2 and 5.2, whose DMD objectives [Yin et al., 2024b,a] commit to a specific sampler at training time; and it is not a model to be driven by a multi-step corrector [??], which would be both redundant—the subtraction in Eq. (48) is already the exact interval integral—and incorrect, since such correctors assume the network predicts the instantaneous velocity v , whereas u_θ predicts the average velocity $u \neq v$.

7.3 Warm-Starting Without Loss

The degenerate diagonal. MeanFlow contains flow matching as its boundary case: when $r = t$, Eq. (44) reduces to $u(x_t, t, t) = v(x_t, t)$ and the target in Eq. (46) degenerates exactly to the flow-matching target of Eq. (43). The converged multi-step teacher therefore already solves the MeanFlow objective on the $r = t$ diagonal of the (r, t) plane, and distillation amounts to extending a function the teacher has learned on a line to the triangle above it—not to training a different model that must first re-approach the teacher.

Table 2: Quantitative comparison of baseline methods. ViCLIP and Self-CIDS assess cross-shot visual and identity consistency, Voice assesses speaker consistency, and Recall measures speech-content coverage. All metrics are higher-is-better. Boldface and underlining indicate the best and second-best results, respectively.

Method	Cross-Shot Consistency			Video Quality		Text Consistency	Speech
	ViCLIP $\uparrow$	Self-CIDS $\uparrow$	Voice $\uparrow$	Aesthetic $\uparrow$	Imaging $\uparrow$	CLIP $\uparrow$	Recall $\uparrow$
JavisDiT++ [Liu et al., 2026]	0.6955	0.5621	0.7933	0.5019	0.6084	0.2522	0.0951
LTX-2.3 [HaCohen et al., 2026]	0.7047	0.6135	0.6847	0.5436	0.4751	0.2559	0.9304
LTX-2.5 [HaCohen et al., 2026]	0.7775	0.6640	0.7603	0.6019	0.5804	<u>0.2683</u>	<u>0.9545</u>
ShotStream+MMAudio [Luo et al., 2026]	0.7988	0.7308	0.7945	0.5255	0.6044	0.1998	0.0297
StoryMem+MMAudio [Zhang et al., 2025b]	0.7897	0.7491	0.7643	0.5265	0.6320	0.2368	0.0301
HappyOyster (Directing)	0.7535	0.6940	0.7705	0.5606	0.6701	0.2544	0.1085
JoyAI-Echo-1.0 [Li et al., 2026]	<u>0.8026</u>	<u>0.7793</u>	<u>0.8129</u>	0.5679	<u>0.7058</u>	0.2658	0.9489
JoyAI-Echo-1.5	0.8264	0.7937	0.8524	<u>0.5705</u>	0.7467	0.2868	0.9674

Zero-initialized interval branch. To realize this extension without disturbing the teacher, the student architecture augments the backbone with a second conditioning branch that mirrors the timestep pathway: a sinusoidal embedding of the interval length $t - r$, passed through its own MLP and its own modulation projection, whose outputs are *added* to the corresponding timestep-branch activations. Feeding $t - r$ rather than r makes the degenerate case correspond to a fixed zero input, and keeping the projections separate—each branch projected before summation—decouples the new branch’s gradients from the pretrained modulation weights. The terminal linear layer of each new pathway is initialized to zero, so at step 0 the branch contributes exactly nothing and the student’s forward pass reproduces the teacher’s by construction; a start-of-training self-check verifies that the loss pipeline degenerates to flow matching when $r = t$, confirming the wiring. The warm start is thus lossless: no adapter distillation, no feature matching, and no re-convergence phase are needed.

Curriculum and confound control. Distillation is full-parameter and inherits the teacher’s entire conditioning stack unchanged—the 36-channel concatenation, the condition noising of Eq. (42), the degradation pipeline, the paired HQ/LQ data, and the per-stage resolution buckets—so that any behavioral difference between student and teacher is attributable to the objective rather than to a data-distribution shift. Training proceeds in two stages: a first stage on moderate-resolution buckets initialized from the converged teacher, followed by a continuation on the teacher’s own high-resolution multi-aspect buckets with the optimizer state carried over; the shape-invariant tangent normalization of Sec. 7.2 is what allows this resolution change without retuning the loss weighting. The optimizer follows the teacher’s settings except for a raised second-moment coefficient (β_2 from 0.95 to 0.99), accommodating the higher gradient variance of the finite-difference target, and the high-resolution stage adds a cross-rank finiteness gate that skips optimizer steps on non-finite gradients before clipping, preserving the Adam state through transient instabilities and closing a failure mode in which a NaN gradient silently bypasses standard norm clipping because the comparison `nan < 1.0` evaluates to false. Combined with the tangent warmup, the overall trajectory is a continuous deformation of the teacher: the student begins as the teacher, is annealed onto the MeanFlow objective, and ends as a single set of weights for which Eq. (48) holds at every step budget.

8 Results

We evaluate JoyAI-Echo-1.5 from two complementary perspectives: *long-horizon audio-visual generation* and *action-conditioned world modeling*. For world modeling, we report results on two public benchmarks, WBench and SANA-WM-Bench, together with a large-scale human preference study.

8.1 Long-Horizon Audio-Visual Generation

Experimental setup. For quantitative comparison, we evaluate long-horizon audio-visual generation on a curated benchmark of 100 stories and 3,000 shots, following the evaluation protocol of JoyAI-Echo-1.0 [Li et al., 2026]. Specifically, our evaluation pipeline covers four complementary dimensions. **(i) Cross-Shot Consistency.** (1) *ViCLIP Similarity:* We compute the pairwise cosine similarity between ViCLIP [Wang et al.] video embeddings across all shot pairs to measure cross-shot visual-

semantic coherence. (2) *Self-CIDS*: We evaluate character identity consistency on a per-role basis across shots, using GroundingDINO [Liu et al., 2023] for character detection, Torchreid [Zhou and Xiang, 2019] for body embeddings, and FaceNet [Schroff et al., 2015] for face embeddings. (3) *Voice Consistency*: We evaluate speaker identity consistency for the same role across shots using the cosine similarity between 3DSpeaker [Chen et al., 2025] speaker-verification embeddings. (ii) **Video Quality**. We adopt the slow-fast evaluation protocol of VBench-Long [Huang et al., 2024]. Specifically, we concatenate the generated shots into a long video and divide it into 2-second clips, on which we evaluate aesthetic quality and imaging quality. (iii) **Text Consistency**. We evaluate the alignment between the generated visual content and the input story script using the *CLIP score* [Radford et al., 2021]. (iv) **Speech Content Fidelity**. We evaluate how faithfully the generated speech covers the scripted dialogue. Specifically, we first transcribe the generated audio using Whisper ASR [Radford et al., 2022], and then compute word-level speech recall against the reference dialogue as the number of matched reference words divided by the total number of reference words. Compared with the accuracy metric used in JoyAI-Echo-1.0, we adopt this recall-based metric and re-evaluate all methods under the same protocol. This metric directly measures the coverage of intended dialogue content, and is particularly useful for distinguishing joint audio-visual generation models that generate meaningful scripted speech from video-to-audio cascade methods that may produce temporally aligned audio without preserving the intended lexical content.

We compare JoyAI-Echo-1.5 against three categories of baselines covering different paradigms of audio-visual generation. **Joint audio-visual generation** baselines include *JavisDiT++* [Liu et al., 2026], a joint audio-visual DiT model with hierarchical spatio-temporal prior synchronization and modality-specific MoE modules, and the *LTX* series [HaCohen et al., 2026], represented by LTX-2.3 and LTX-2.5, which jointly generate synchronized audio and video with Gemma-based prompt enhancement. For these methods, each shot is generated from its corresponding structured prompt with a duration of approximately 10 seconds, and the resulting shots are concatenated for long-form evaluation. For LTX-2.3 and LTX-2.5, we enable the built-in Gemma prompt-enhancement module and, from the second shot onward, condition each generation on the final frame of the preceding shot to propagate local visual continuity across adjacent shots. **Cascade generation** baselines include *ShotStream* [Luo et al., 2026]+*MMAudio* [Cheng et al., 2025] and *StoryMem* [Zhang et al., 2025b]+*MMAudio* [Cheng et al., 2025]. Under the same 30-shot setting, ShotStream and StoryMem first generate a 10-second silent video for each shot from its visual prompt, after which MMAudio synthesizes the corresponding audio using shot-level audio descriptions when applicable; all generated audio-visual shots are then concatenated into a single long video. This protocol ensures that these baselines are evaluated under the same story scripts, shot durations, and final long-form format. Finally, we compare with **native long-form audio-visual generation** using *Happy Oyster*, a recently released streaming generative world model that natively produces synchronized visual and auditory signals. We use its *Directing mode*,¹ which takes a narrative script as input and continuously generates a coherent audio-visual sequence while maintaining scene continuity over time. Because Happy Oyster supports at most 3 minutes of continuous generation, we evaluate it using the scripts corresponding to the first 18 shots of each story.

Quantitative Comparison. Table 2 compares JoyAI-Echo-1.5 with native audio-visual generators, cascaded long-video systems, and JoyAI-Echo-1.0. The evaluation covers four complementary aspects: cross-shot consistency, local video quality, text–video alignment, and speech-content fidelity. JoyAI-Echo-1.5 achieves the best result on six of the seven reported metrics and ranks second on the remaining aesthetic quality metric. Importantly, the improvements are obtained simultaneously across visual, acoustic, and semantic dimensions, rather than by optimizing one modality at the expense of another.

For cross-shot consistency, JoyAI-Echo-1.5 obtains ViCLIP, Self-CIDS, and voice-consistency scores of 0.8264, 0.7937, and 0.8524, respectively, outperforming all competing methods. Compared with JoyAI-Echo-1.0, these scores increase by 0.0238, 0.0144, and 0.0395. The particularly pronounced improvement in voice consistency indicates that the model preserves speaker characteristics more reliably over temporally separated shots. We attribute this gain to the unified audio-visual Memory representation: visual observations provide recurring appearance cues, while speech-filtered audio Memory supplies cleaner speaker information. Their joint use allows the model to retrieve character appearance and voice characteristics through the same conditional interface. The separated RoPE

¹Details available at <https://www.happyoyster.cn/docs>

coordinates further distinguish multiple Memory sources from the current prediction target, enabling composable conditioning without imposing the slot-specific attention masks used in JoyAI-Echo-1.0.

The visual-consistency results also demonstrate the importance of learning from scene-disjoint Memory–target pairs. Although cascaded systems such as ShotStream+MMAudio and StoryMem+MMAudio obtain competitive ViCLIP and Self-CIDS scores, JoyAI-Echo-1.5 performs better on both metrics while also maintaining substantially stronger speaker consistency. This suggests that the model does not merely preserve coarse scene semantics across shots, but more reliably maintains the visual identity and acoustic identity of recurring characters. Compared with LTX-2.5, which achieves strong independent-shot generation quality, JoyAI-Echo-1.5 also exhibits a clear advantage on all three cross-shot metrics.

In addition to long-range consistency, JoyAI-Echo-1.5 improves local generation fidelity. It achieves the highest imaging-quality score of 0.7467, an absolute improvement of 0.0409 over JoyAI-Echo-1.0, while attaining the second-best aesthetic score of 0.5705. These results indicate that introducing Memory conditioning does not constrain the model to low-diversity or identity-centric training samples. The second-stage data curriculum, which combines Memory data with high-quality subtitle-free videos at multiple resolutions and aspect ratios, preserves the general visual-generation capability of the underlying model. The increased imaging score is consistent with the improved structural clarity, color balance, and rendering stability observed in the qualitative examples.

JoyAI-Echo-1.5 further reaches the best CLIP score of 0.2868, improving over JoyAI-Echo-1.0 by 0.0210 and over the strongest competing baseline, LTX-2.5, by 0.0185. This improvement shows that the gains in consistency do not arise from repeatedly generating similar or weakly varying content. Instead, the model remains responsive to shot-specific text instructions while preserving information retrieved from Memory. The transition-aware captions introduced for multi-shot training additionally provide explicit supervision for shot boundaries, newly appearing subjects, camera changes, actions, and audio events, improving the model’s ability to follow temporally structured descriptions.

For speech-content fidelity, JoyAI-Echo-1.5 obtains the highest Recall score of 0.9674. It surpasses JoyAI-Echo-1.0 and LTX-2.5 by 0.0185 and 0.0129, respectively, and substantially outperforms cascaded video-to-audio systems such as ShotStream+MMAudio and StoryMem+MMAudio. The low Recall of these cascaded approaches indicates that generating audio after video synthesis does not reliably preserve the scripted dialogue. By contrast, JoyAI-Echo-1.5 jointly predicts audio and video under shared textual and Memory conditions, allowing speech content, speaker characteristics, and visible character motion to be modeled within a unified generation process. Overall, the quantitative results verify the central design objective of JoyAI-Echo-1.5: improving composable Memory conditioning and long-range identity preservation while retaining strong standalone T2AV and I2AV generation quality.

User Study. We further conduct a human preference study to evaluate perceptual properties that are not fully captured by automatic metrics. The study focuses on the long-video variant of JoyAI-Echo-1.5 and compares it with HappyOyster in directing mode. The two generated videos in each pair are presented in randomized and anonymized order. Following the Good–Similar–Bad (GSB) protocol, annotators determine whether the JoyAI-Echo-1.5 result is better, the two results are perceptually similar, or the HappyOyster result is better with respect to a specified evaluation aspect. Each aspect is evaluated independently to prevent overall visual preference from dominating judgments about instruction following, identity consistency, or audio.

As shown in Table 3, JoyAI-Echo-1.5 receives a higher preference rate than HappyOyster across all five evaluated dimensions. The largest margin is observed in audio-visual synchronization: JoyAI-Echo-1.5 is preferred in 48.4% of the comparisons, compared with 20.3% for HappyOyster, corresponding to a 28.1 percentage-point advantage. This result complements the automatic voice-consistency and speech-Recall metrics. It indicates that the generated audio not only preserves speaker identity and dialogue content, but also remains perceptually aligned with visible speaking motion and shot-level events.

JoyAI-Echo-1.5 also shows clear advantages in semantic controllability. It is preferred over HappyOyster by 38.6% versus 17.7% for story-instruction following, and by 39.2% versus 23.5% for dialogue and background-audio instruction following. These two dimensions evaluate different levels of conditioning: the former measures whether the generated sequence realizes the requested characters, actions, scenes, and narrative progression, whereas the latter focuses on spoken content



Figure 3: Qualitative long-horizon generation results for JoyAI-Echo-1.5. The selected frames illustrate stable character appearance, scene continuity, and balanced visual rendering across a multi-shot story.

and the surrounding acoustic environment. The results suggest that joint audio-visual generation provides more faithful control over both narrative events and their associated sounds.

For Memory and identity consistency, JoyAI-Echo-1.5 is preferred in 24.8% of the pairs, compared with 15.7% for HappyOyster. The relatively large Similar proportion of 59.5% indicates that both systems preserve identity adequately in many cases; nevertheless, when a perceptible difference occurs, annotators favor JoyAI-Echo-1.5 more frequently. This result is consistent with its leading ViCLIP, Self-CIDS, and voice-consistency scores, providing human-evaluation evidence that the unified Memory mechanism improves both visual and acoustic identity preservation.

Finally, JoyAI-Echo-1.5 receives a higher preference rate for audio quality (44.4% versus 33.3%). Together with the synchronization and dialogue-following results, this demonstrates that the improvement is not limited to the presence of speech. The model produces audio that is more faithful to the requested content, more closely aligned with the video, and more perceptually favorable over long generation horizons. Collectively, the user study confirms the improvements measured by the automatic evaluation and shows that they remain perceptible in complete long-video outputs.

Qualitative Comparison. Figure 3 presents frames sampled at widely separated timestamps from a long-horizon generation produced by JoyAI-Echo-1.5. The sequence spans multiple indoor and outdoor locations, changes in camera scale, and interactions involving recurring and newly introduced characters. Despite these substantial variations, the principal characters retain recognizable facial structure, hairstyle, clothing-related cues, and overall visual identity across distant shots. The generated sequence also maintains a coherent cinematic appearance, with stable exposure and restrained color variation rather than progressively accumulating oversaturation or contrast artifacts.

These examples illustrate the complementary roles of the model and the long-video generation pipeline. At the generator level, unified audio-visual Memory conditioning provides persistent identity and speaker cues while still allowing each target shot to follow its own textual description. The high-quality training corpus improves local rendering quality and broadens the model’s spatial and linguistic coverage, while transition-aware multi-shot captions strengthen its ability to represent changes in subject, action, camera view, and scene context around shot boundaries. At the system level, the Director Agent selects grounded Memory observations and revises the shot-level plan before subsequent generation, reducing the propagation of incorrect characters, actions, or scene attributes.

Compared with JoyAI-Echo-1.0, JoyAI-Echo-1.5 therefore exhibits improvements at both local and long-range scales. Individual shots show clearer structure and more balanced rendering, while the full sequence exhibits less character drift, scene drift, and color accumulation over time. The qualitative results are consistent with the improvements in imaging quality, CLIP alignment, cross-shot visual consistency, and speaker consistency reported in Table 2. A more complete long-horizon visualization with densely sampled timestamps is provided in the supplementary material.

Table 3: Human evaluation based on Good–Similar–Bad (GSB) pairwise comparisons between JoyAI-Echo-1.5 (long-video variant) and HappyOyster for long-horizon audio-visual generation. Numbers denote the percentage of user preferences and may not sum to exactly 100% because of rounding.

Aspect	JoyAI-Echo-1.5	Similar	HappyOyster
Story-instruction following	38.6%	43.8%	17.7%
Dialogue/background-audio following	39.2%	37.3%	23.5%
Memory/identity consistency	24.8%	59.5%	15.7%
Audio-visual synchronization	48.4%	31.4%	20.3%
Audio quality	44.4%	22.2%	33.3%

Table 4: Comparison on the 158-case WBench Navigation split [Ying et al., 2026]. Results are sorted by Average.

Model	Avg.	Quality	Setting	Interaction	Consistency	Physical
JoyAI-Echo-1.5	81.7	81.5	79.4	87.2	89.8	70.6
HiDream-O1-World [Ying et al., 2026]	80.7	81.9	81.9	79.5	88.0	72.1
LingBot-World (fast v2) [Robbyant Team et al., 2026]	79.4	81.8	76.8	82.8	86.5	69.1
Kling 3.0 [Ying et al., 2026]	79.0	81.4	91.0	69.4	83.7	69.3
LingBot-World (base-camera) [Robbyant Team et al., 2026]	78.5	78.9	72.6	80.1	89.9	71.2
Wan 2.7 [Wan Team et al., 2025]	78.1	81.5	<u>91.4</u>	64.4	81.6	<u>71.8</u>
HY-World 1.5 (AR-distill) [Sun et al., 2025]	78.1	78.1	<u>72.2</u>	<u>86.8</u>	86.9	<u>66.3</u>
HY-Video 1.5 [Wu et al., 2025]	77.9	77.6	85.6	71.4	87.4	67.4
LingBot-World (fast) [Robbyant Team et al., 2026]	77.4	79.4	77.9	79.2	84.9	65.7
HappyOyster [Ying et al., 2026]	76.8	77.3	74.2	84.9	84.3	63.5
Lyra 2.0 (4-step AR) [Shen et al., 2026]	76.4	77.1	73.2	85.6	79.3	66.7
Seedance 1.5 [Seedance et al., 2025]	76.2	82.1	82.9	66.3	81.3	68.4
Cosmos3-Super [NVIDIA, 2026]	76.0	76.4	91.6	58.0	85.0	69.2
SANA-WM (4-step AR) [Zhu et al., 2026]	76.0	79.3	76.1	82.2	80.7	61.9
DreamX-World (5B AR) [DreamX Team et al., 2026]	75.0	77.5	80.8	78.6	74.9	63.3
ABot-World [Jiang et al., 2026]	74.7	76.8	71.4	84.0	79.5	61.7
Cosmos 2.5 [NVIDIA, 2025]	74.6	72.9	83.3	63.1	86.5	67.4
Cosmos3-Nano [NVIDIA, 2026]	74.2	77.4	87.3	59.1	83.7	63.6
LTX-2.3 [HaCohen et al., 2026]	74.2	77.1	85.2	66.4	77.2	64.9
Genie 3 [Google DeepMind, 2025]	73.9	75.2	72.5	73.4	82.6	65.7

8.2 Action-Conditioned World Modeling

We evaluate interactive world modeling on two public benchmarks, WBench [Ying et al., 2026] and SANA-WM-Bench [Zhu et al., 2026]. The former evaluates broad interactive-world behavior, while the latter focuses on camera-trajectory following and long-horizon visual persistence. We evaluate the **world-model** variant of JoyAI-Echo-1.5 in this subsection.

WBench. We evaluate JoyAI-Echo-1.5 on the official 158-case Navigation split of WBench, covering both first- and third-person interaction. We follow the official multi-turn protocol and report the five evaluation dimensions—Quality, Setting, Interaction, Consistency, and Physical—as well as their overall Average.

As shown in Table 4, JoyAI-Echo-1.5 ranks first among the 31 evaluated models with an Average score of 81.7, outperforming the second-ranked HiDream-O1-World by 1.0 point. It achieves the highest Interaction score of 87.2 and the second-highest Consistency score of 89.8, showing strong action adherence together with stable world evolution across multiple interaction turns. Its Quality, Setting, and Physical scores are 81.5, 79.4, and 70.6, respectively.

Qualitative comparison. We further provide qualitative comparisons on representative WBench Navigation cases in Figure 4. Each example shows temporally ordered generations under the same initial observation and navigation condition, allowing direct comparison of how different models respond to multi-step control. Several consistent trends can be observed. Strong world modeling requires more than producing visually plausible individual frames: the model must preserve the identity and structure of the initial world while evolving the viewpoint according to the navigation input. Across the shown examples, competing methods often exhibit one or more failure modes, including abrupt viewpoint jumps, unintended zoom-in behavior, large changes in subject scale, loss of scene layout, and drift toward visually unrelated content. These errors accumulate over time and

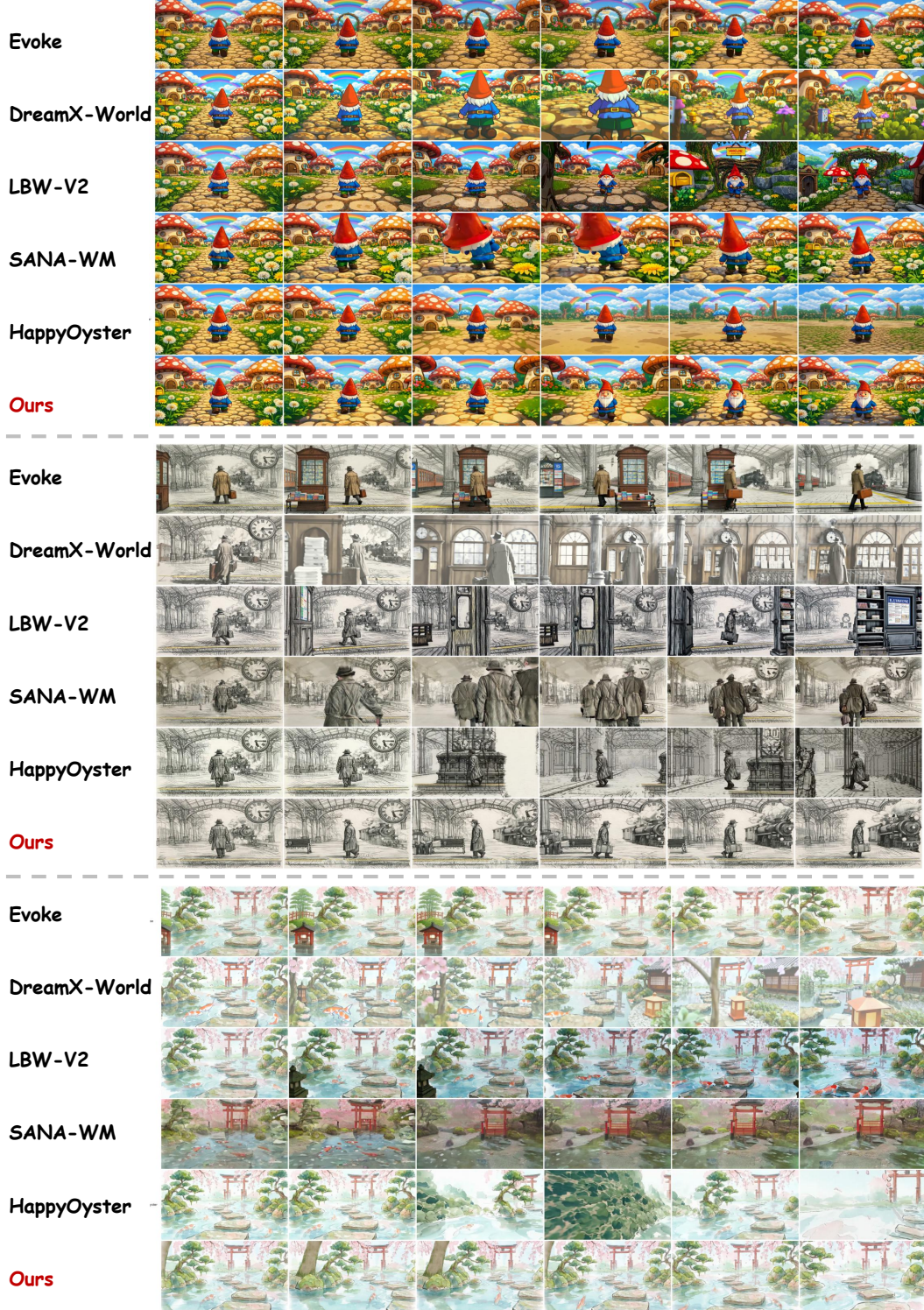


Figure 4: Qualitative comparison on representative WBench Navigation cases across diverse visual domains. Each row shows temporally ordered generations under the same initial observation and navigation condition. Compared with competing methods, our method better preserves scene identity and structure while maintaining more coherent viewpoint evolution over multiple interaction steps.

Table 5: Short-horizon comparison on SANA-WM-Bench using the front 241 frames of each trajectory. Each split contains 80 scenes. VBench Overall follows the benchmark convention of reporting normalized visual quality on a 0–100 scale. Camera accuracy reports rotation error R in degrees, relative translation error T , and camera-motion consistency error CMC after similarity alignment.

Method	VBench Overall $\uparrow$	R $\downarrow$	T $\downarrow$	CMC $\downarrow$
<i>Simple-Trajectory Split</i>				
Matrix-Game 3 [Wang et al., 2026]	79.41	4.474	0.406	0.453
LingBot-World-v2 [Robbyant Team et al., 2026]	81.13	3.092	0.282	0.310
HY-WorldPlay [Sun et al., 2025]	82.05	2.256	0.284	0.298
DreamX-World [DreamX Team et al., 2026]	<u>82.42</u>	1.839	0.261	0.273
SANA-WM [Zhu et al., 2026]	81.75	0.489	0.194	<u>0.195</u>
JoyAI-Echo-1.5	83.91	<u>0.523</u>	0.160	0.162
<i>Hard-Trajectory Split</i>				
Matrix-Game 3 [Wang et al., 2026]	79.37	12.819	0.583	0.717
LingBot-World-v2 [Robbyant Team et al., 2026]	81.67	6.367	0.357	0.426
HY-WorldPlay [Sun et al., 2025]	81.24	6.346	0.387	0.437
DreamX-World [DreamX Team et al., 2026]	<u>82.03</u>	8.804	0.467	0.556
SANA-WM [Zhu et al., 2026]	81.79	1.248	0.199	0.206
JoyAI-Echo-1.5	83.96	<u>1.697</u>	<u>0.256</u>	<u>0.266</u>

become particularly visible in longer rollouts, where local frame quality alone is insufficient to ensure stable interaction. In contrast, JoyAI-Echo-1.5 produces more stable multi-turn evolution across diverse visual domains, including stylized scenes, sketch-like environments, natural outdoor scenes, indoor tabletop settings, and structured sci-fi corridors. The generated rollouts better preserve scene identity, maintain more coherent camera progression, and exhibit fewer abrupt structural changes across steps. This advantage is especially clear in scenes with repeated spatial patterns or strong geometric constraints, where inaccurate control often leads to rapid layout drift in baseline methods. These qualitative comparisons are consistent with the quantitative benchmark results. The strong WBench Interaction and Consistency scores of JoyAI-Echo-1.5 are reflected visually in its improved control adherence and world stability: the model not only responds to the intended navigation signal, but also maintains a more coherent and persistent visual world throughout the rollout.

SANA-WM-Bench. We further evaluate trajectory control and long-horizon persistence on SANA-WM-Bench. Following its official protocol, we report rotation error (R), relative translation error (T), camera-motion consistency error (CMC), and VBench Overall. We consider both a 241-frame setting for short-horizon generation and a 961-frame setting for long-horizon chained generation. We retain the checkpoint names used in each evaluation: the 241-frame setting uses LingBot-World-v2, whereas the published 961-frame comparison reports LingBot-World.

At 241 frames, JoyAI-Echo-1.5 achieves the highest VBench Overall on both trajectory splits, reaching 83.91 on Simple and 83.96 on Hard. On Simple trajectories, it also obtains the lowest translation and CMC errors, while maintaining rotation accuracy close to SANA-WM. On Hard trajectories, SANA-WM retains stronger pose accuracy, whereas JoyAI-Echo-1.5 preserves the best visual quality. For the 961-frame long-horizon setting, JoyAI-Echo-1.5 achieves a VBench Overall of 81.36 on the Simple split, together with the highest revisit PSNR of 15.10 dB and the lowest translation and CMC errors among the reported chained-generation results. Performance remains visually stable over long horizons, although rotation error increases from 0.523° at 241 frames to 3.22° over the complete Simple trajectory and reaches 12.05° on the Hard split. These results suggest that accumulated rotational drift under difficult long-horizon trajectories remains the primary limitation.

User study. We additionally conduct a subjective evaluation on a self-built World-Model Benchmark (WMB) containing 200 cases. The benchmark covers game, humanoid-robot, natural outdoor, household indoor, and driving scenes, with 132 third-person and 68 first-person cases. We compare JoyAI-Echo-1.5 with LingBot-World-v2 [Robbyant Team et al., 2026] and HappyOyster through randomized and anonymized side-by-side comparisons. Annotators evaluate semantic following, initial-world consistency, spatiotemporal consistency, motion quality, and visual aesthetics, together with an overall preference. Against LingBot-World-v2, JoyAI-Echo-1.5 receives higher explicit

Table 6: Long-horizon comparison on the full SANA-WM-Bench trajectory. Revisit consistency is measured on same-pose pairs. VBench Overall is the normalized Quality component. Temporal degradation is $\Delta IQ = IQ_{0-10} - IQ_{50-60}$ in percentage points; lower is better, and a negative value indicates improvement in the last window. Best and second-best results within each split are shown in bold and underlined, respectively.

Method	Trajectory and visual quality				Revisit consistency			Temporal
	VBench $\uparrow$	$R \downarrow$	$T \downarrow$	CMC $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	$\Delta IQ \downarrow$
<i>Simple-Trajectory Split</i>								
Infinite-World [Wu et al., 2026]	79.18	16.55	1.98	2.08	12.60	0.284	0.595	6.72
LingBot-World [Robbyant Team et al., 2026]	81.82	10.47	2.01	2.05	<u>14.59</u>	0.366	0.394	<u>0.04</u>
HY-WorldPlay [Sun et al., 2025]	68.82	17.89	2.36	2.45	12.83	0.321	0.616	23.59
Matrix-Game 3 [Wang et al., 2026]	78.53	12.96	1.83	1.92	12.29	0.326	0.553	2.41
SANA-WM [Zhu et al., 2026]	79.29	<u>7.59</u>	<u>1.59</u>	<u>1.63</u>	14.16	0.333	0.458	3.79
JoyAI-Echo-1.5	<u>81.36</u>	3.22	0.918	0.927	15.10	<u>0.335</u>	<u>0.451</u>	0.02
<i>Hard-Trajectory Split</i>								
Infinite-World [Wu et al., 2026]	79.51	41.31	2.49	2.84	12.04	0.248	0.617	4.16
LingBot-World [Robbyant Team et al., 2026]	81.89	18.99	<u>1.65</u>	1.81	14.08	<u>0.332</u>	0.436	<u>0.58</u>
HY-WorldPlay [Sun et al., 2025]	70.46	35.46	2.34	2.64	13.72	0.328	0.654	25.88
Matrix-Game 3 [Wang et al., 2026]	78.79	18.79	1.67	1.82	12.17	0.317	0.556	0.32
SANA-WM [Zhu et al., 2026]	79.60	10.02	1.66	<u>1.72</u>	<u>14.10</u>	0.327	<u>0.469</u>	3.09
JoyAI-Echo-1.5	<u>81.72</u>	<u>12.05</u>	1.265	1.358	14.58	0.333	0.482	0.93

Table 7: Human evaluation based on Good–Similar–Bad (GSB) pairwise comparisons between JoyAI-Echo-1.5 (world model variant) and Happy Oyster for world modeling. Numbers denote the percentage of user preferences.

Aspect	JoyAI-Echo-1.5	Similar	Happy Oyster
Semantic following	13.63%	56.94%	29.44%
Initial-world consistency	39.50%	45.06%	15.44%
Motion	18.94%	56.51%	24.56%
Spatiotemporal consistency	57.69%	31.88%	10.44%
Visual aesthetics	52.88%	23.32%	23.81%
Overall	63.13%	9.82%	27.06%

preference across all five dimensions, with the largest margins in semantic following (34.0% vs. 15.1%) and spatiotemporal consistency (32.6% vs. 14.7%). The overall preference is 46.5% for JoyAI-Echo-1.5 versus 21.6% for LingBot-World-v2. Against HappyOyster, JoyAI-Echo-1.5 shows substantial advantages in spatiotemporal consistency (57.7% vs. 10.4%), visual aesthetics (52.9% vs. 23.8%), and initial-world consistency (39.5% vs. 15.4%). HappyOyster is preferred more often in semantic following and motion quality, while JoyAI-Echo-1.5 achieves a substantially higher overall preference (63.1% vs. 27.1%). These results complement the public benchmarks, showing that the gains in visual quality and spatiotemporal consistency are also reflected in human preference.

9 Conclusion

We presented a framework for extending audio-visual generation beyond isolated clips toward persistent stories and interactive worlds. The long-video system uses composable cross-shot audio-visual memory to preserve character appearance, speaker identity, and narrative continuity under flexible text, image, and memory conditioning. The world-model system translates heterogeneous navigation inputs into calibrated metric 6-DoF trajectories, enabling precise and controller-agnostic interaction across diverse environments and viewpoints. Together with few-step distillation, causal audio-visual teacher forcing, and short- and long-horizon Self-Gradient Forcing, these designs allow the models to generate synchronized audio and video efficiently while remaining stable under self-generated histories. Experiments demonstrate improved cross-shot consistency, instruction following, speech fidelity, trajectory control, and long-horizon world persistence, with leading results on both long-form generation evaluations and public world-model benchmarks. These findings suggest that memory, geometric control, and rollout-aware training are complementary foundations for video systems that can remember past events, respond to user intent, and continue evolving over time. Accumulated rotational drift on difficult long-horizon trajectories remains a key limitation, motivating

Table 8: Human evaluation based on Good–Similar–Bad (GSB) pairwise comparisons between JoyAI-Echo-1.5 (world model variant) and LingBot-World-v2 for world modeling. Numbers denote the percentage of user preferences.

Aspect	JoyAI-Echo-1.5	Similar	LingBot-World-v2
Semantic following	34.00%	50.86%	15.14%
Initial-world consistency	14.64%	77.65%	7.71%
Motion	25.29%	54.71%	20.00%
Spatiotemporal consistency	32.64%	52.65%	14.71%
Visual aesthetics	32.50%	47.57%	19.93%
Overall	46.50%	31.92%	21.57%

future work on stronger geometric consistency, persistent world-state representations, and more reliable open-ended interaction.

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